

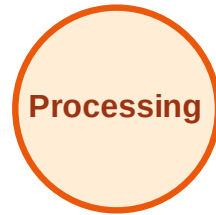
BFS Traversal

Step 1: Start at A (Seeking target node K)

Legend



Complete



Processing

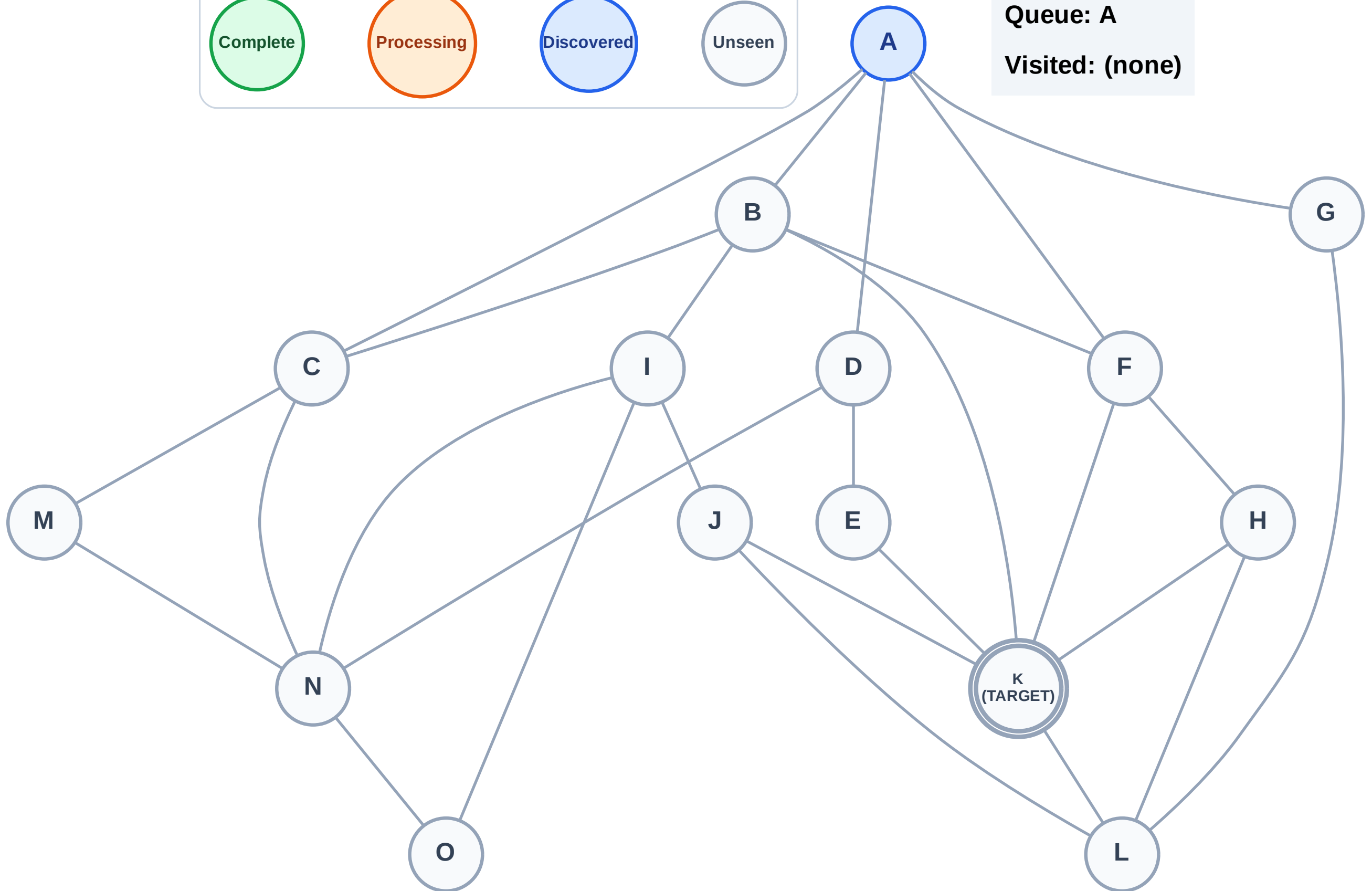


Discovered



Unseen

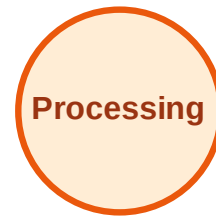
Queue: A
Visited: (none)



BFS Traversal

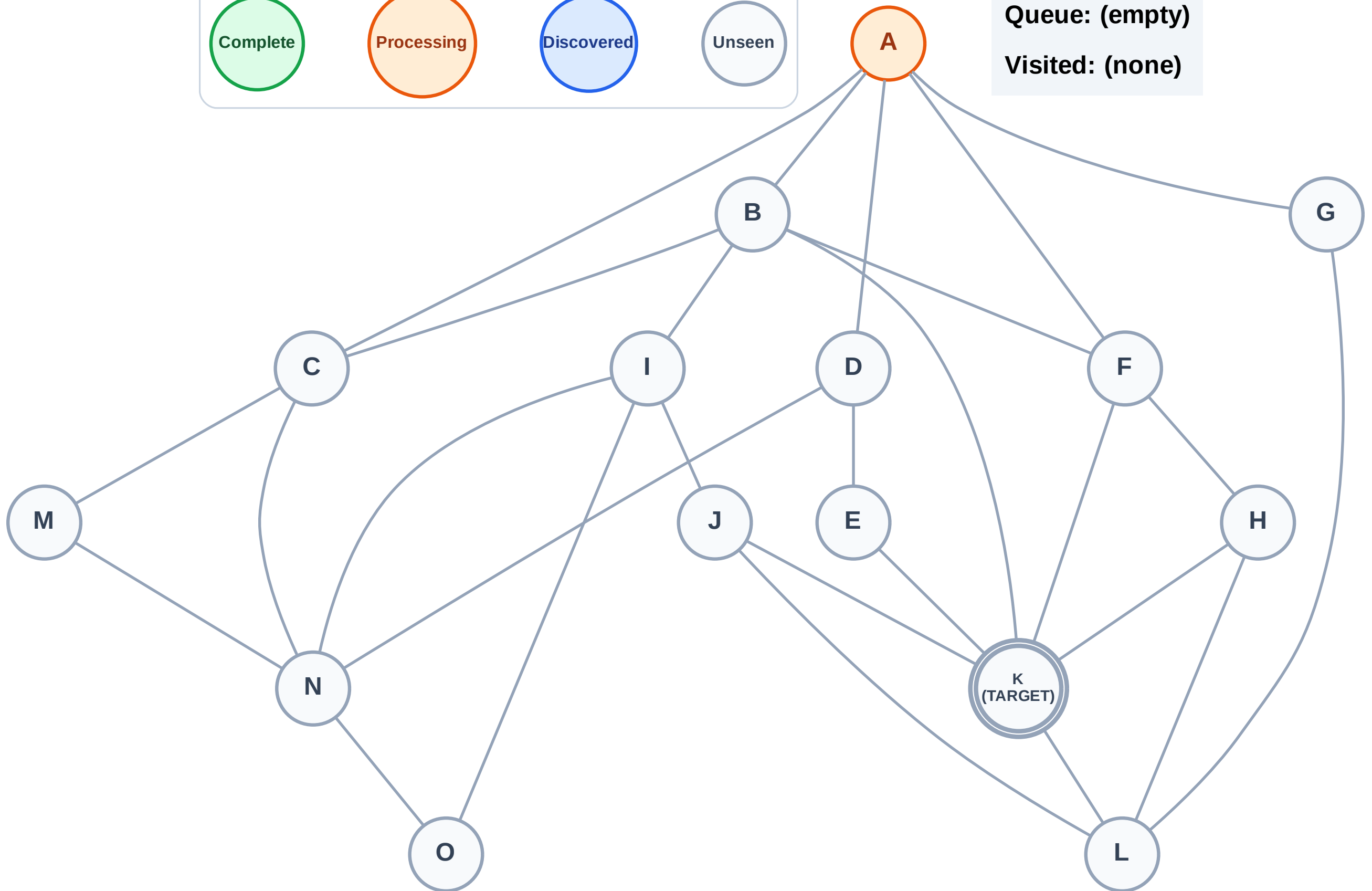
Step 2: Process node A

Legend



Queue: (empty)

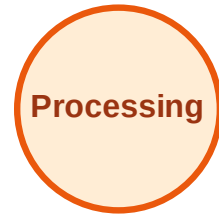
Visited: (none)



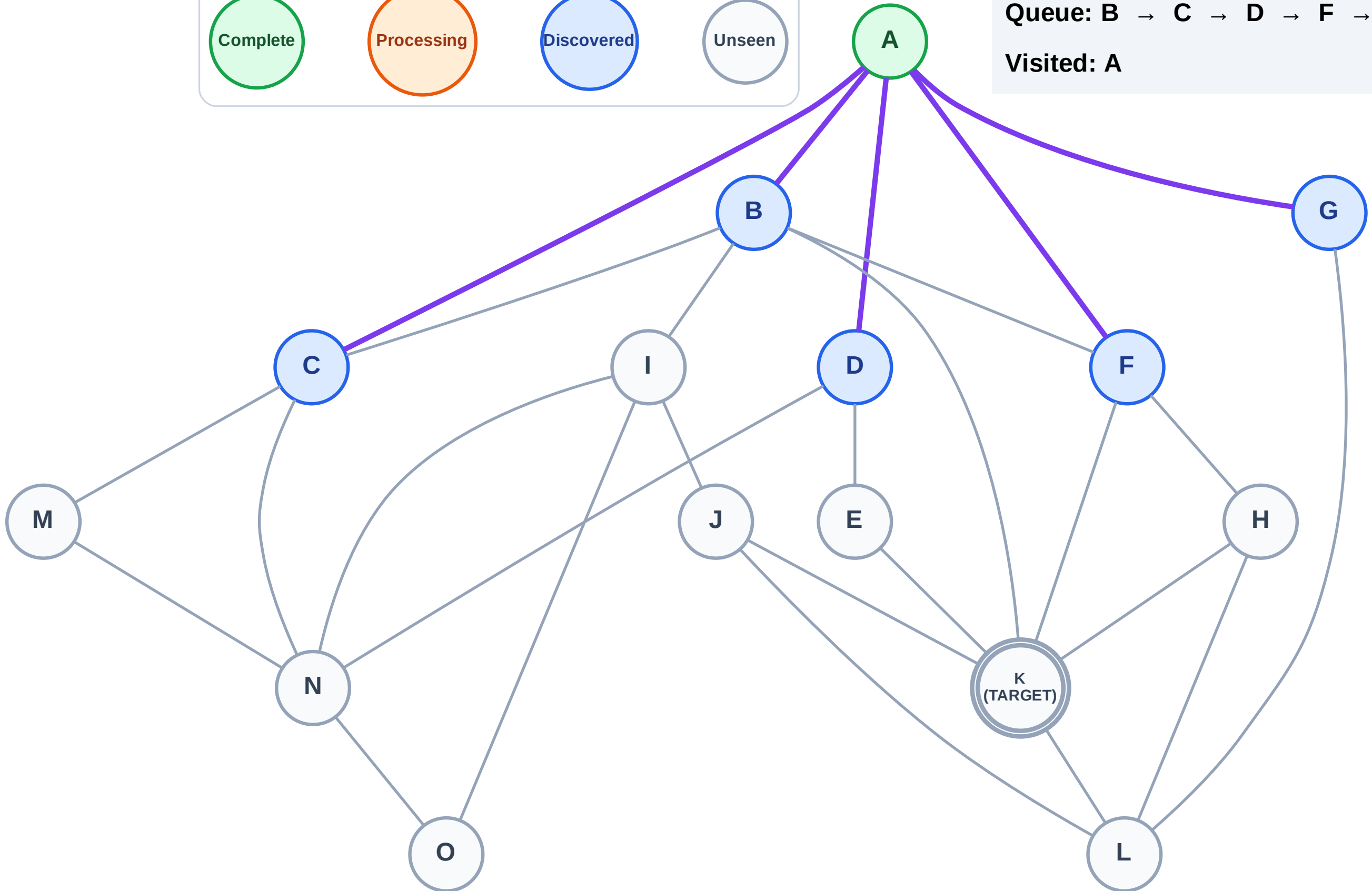
BFS Traversal

Step 3: Discover B, C, D, F, G from A

Legend



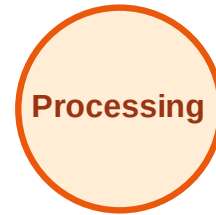
Queue: B → C → D → F → G
Visited: A



BFS Traversal

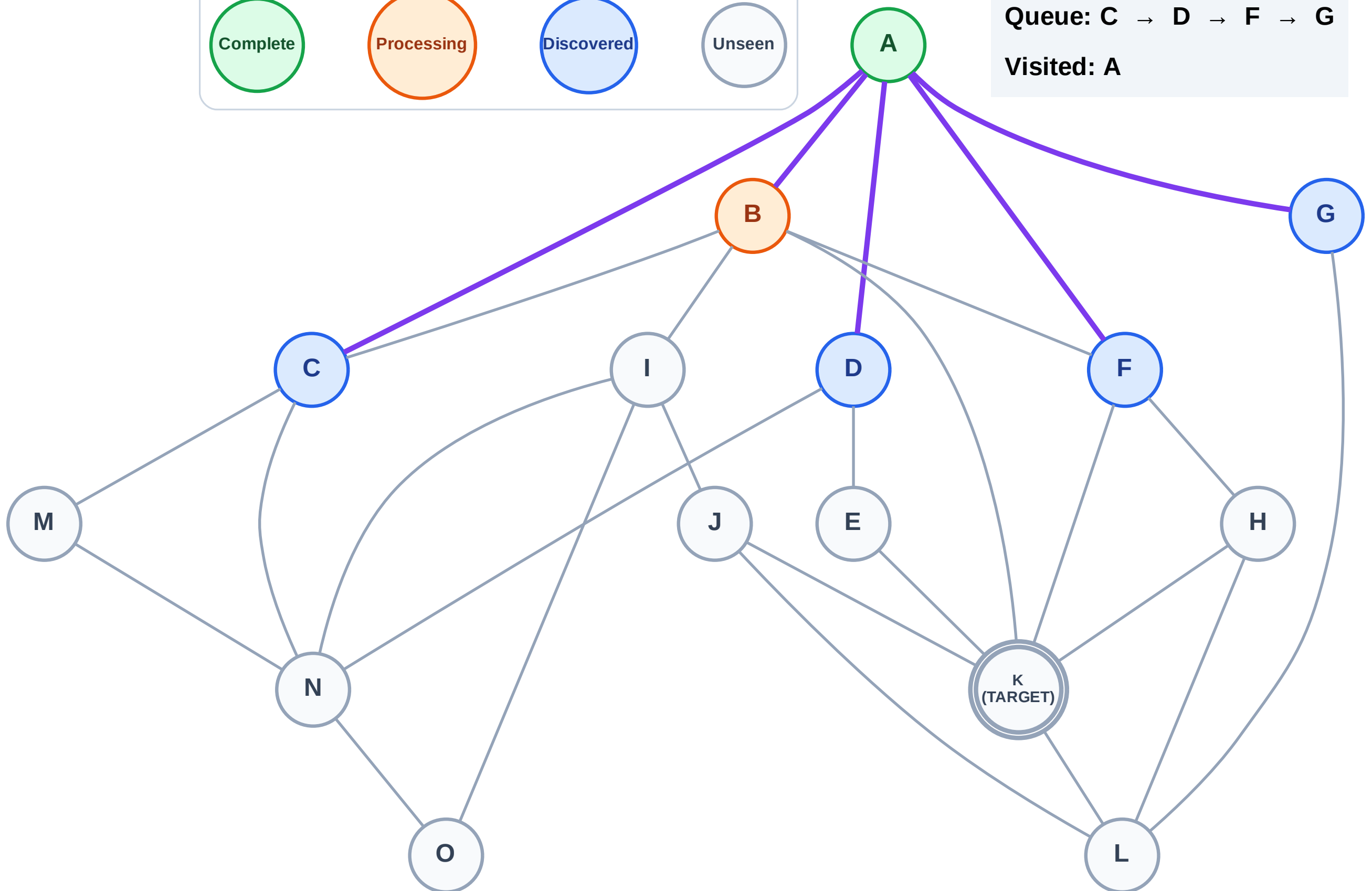
Step 4: Process node B

Legend



Queue: C → D → F → G

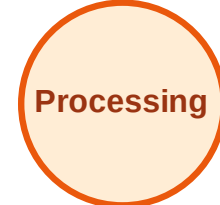
Visited: A



BFS Traversal

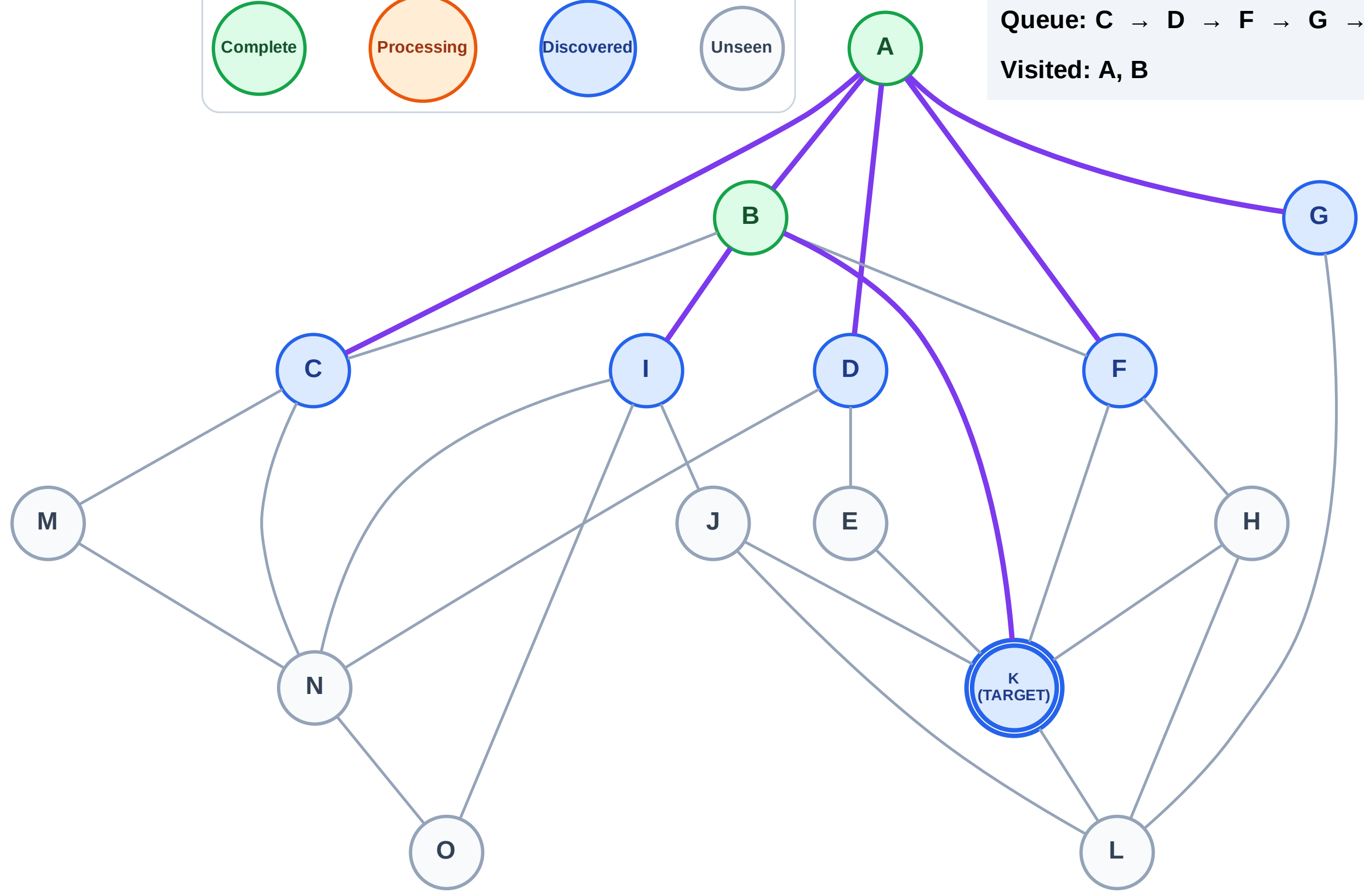
Step 5: Discover I, K from B

Legend



Queue: C → D → F → G → I → K

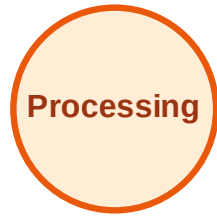
Visited: A, B



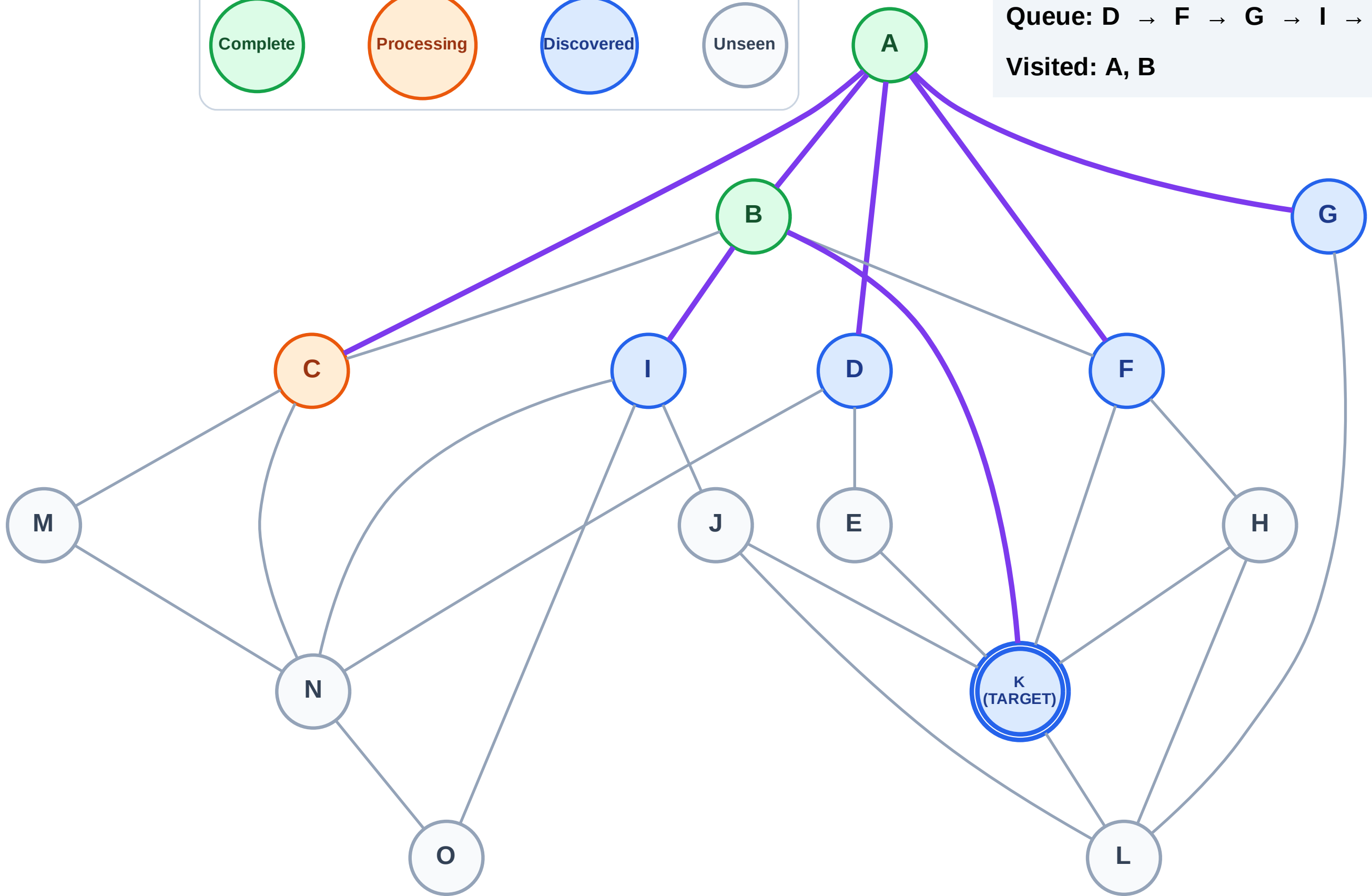
BFS Traversal

Step 6: Process node C

Legend



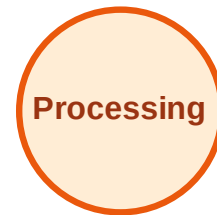
Queue: D → F → G → I → K
Visited: A, B



BFS Traversal

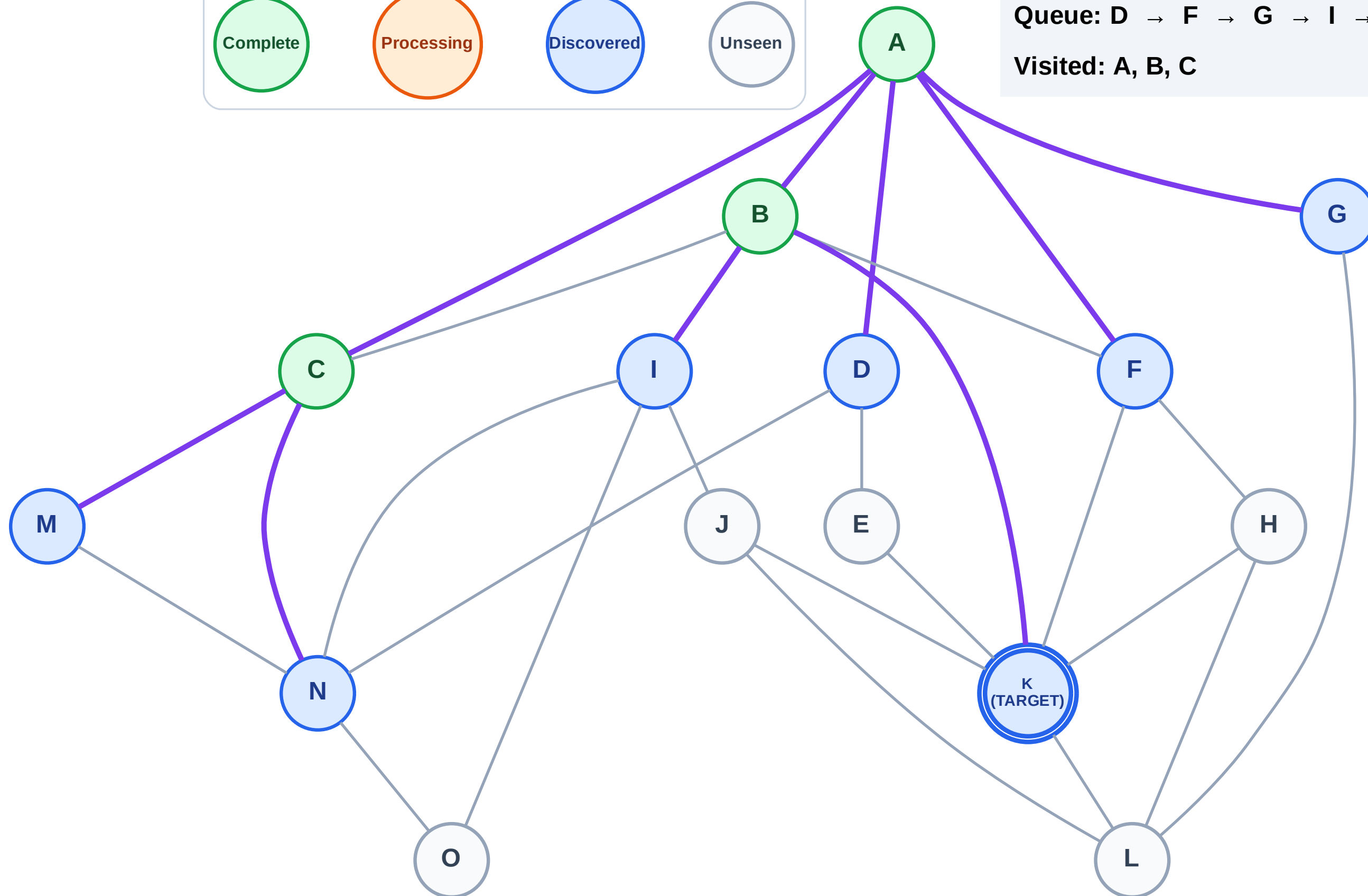
Step 7: Discover M, N from C

Legend



Queue: D → F → G → I → K → M → N

Visited: A, B, C



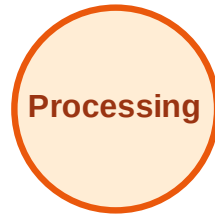
BFS Traversal

Step 8: Process node D

Legend



Complete



Processing



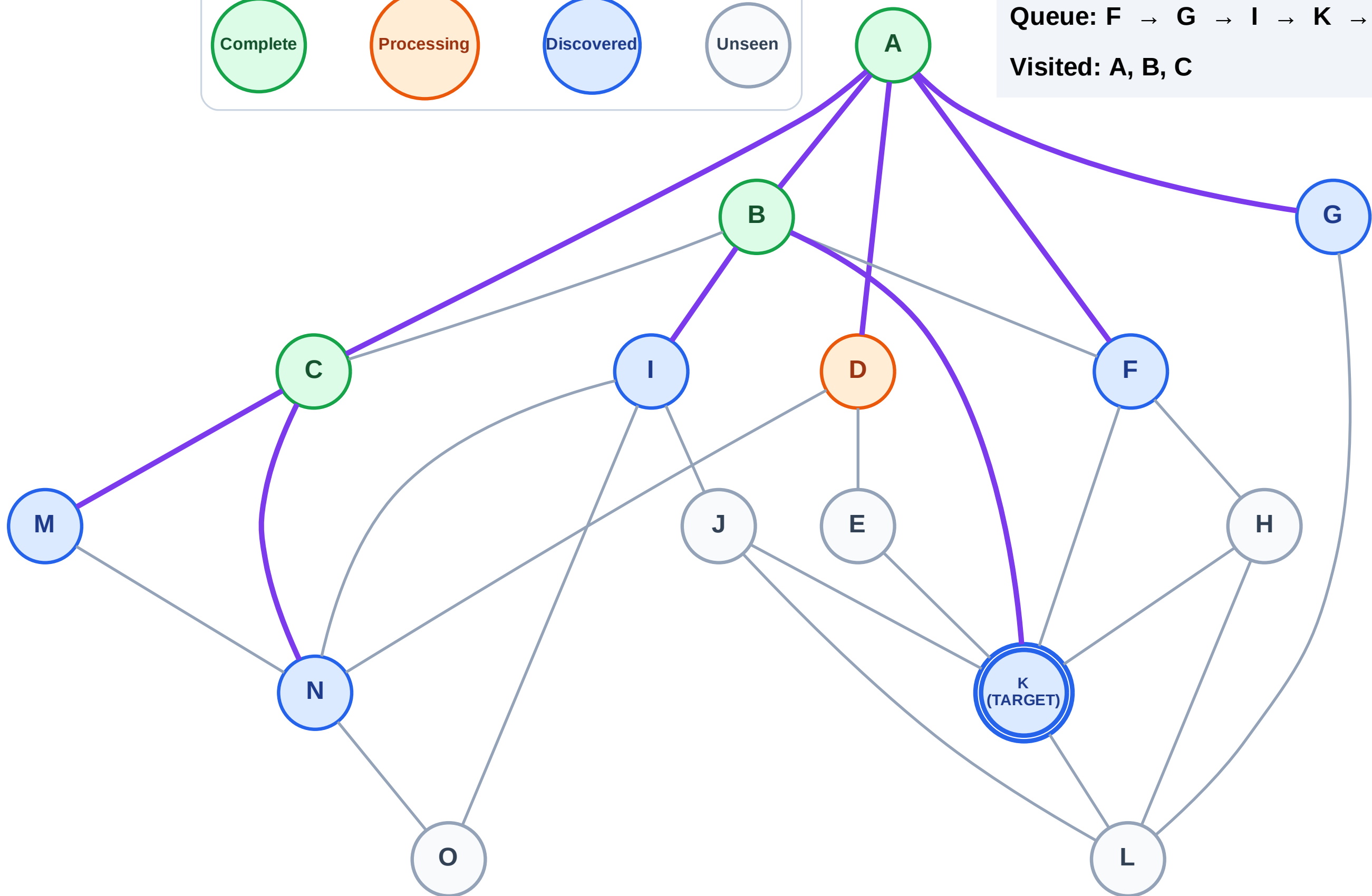
Discovered



Unseen

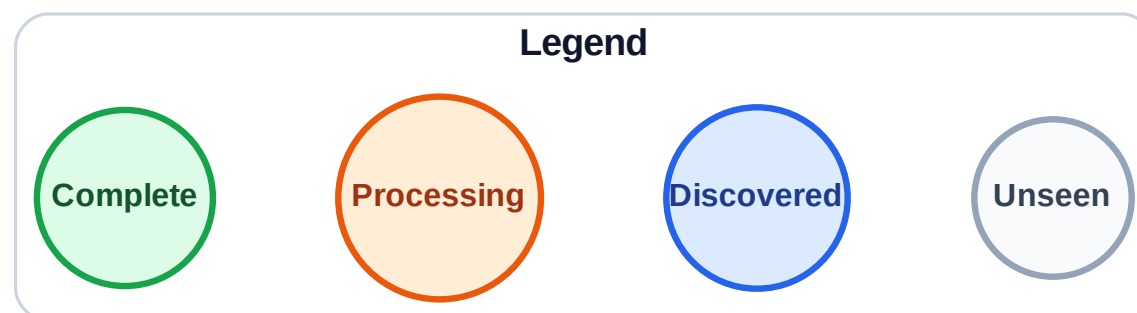
Queue: F → G → I → K → M → N

Visited: A, B, C



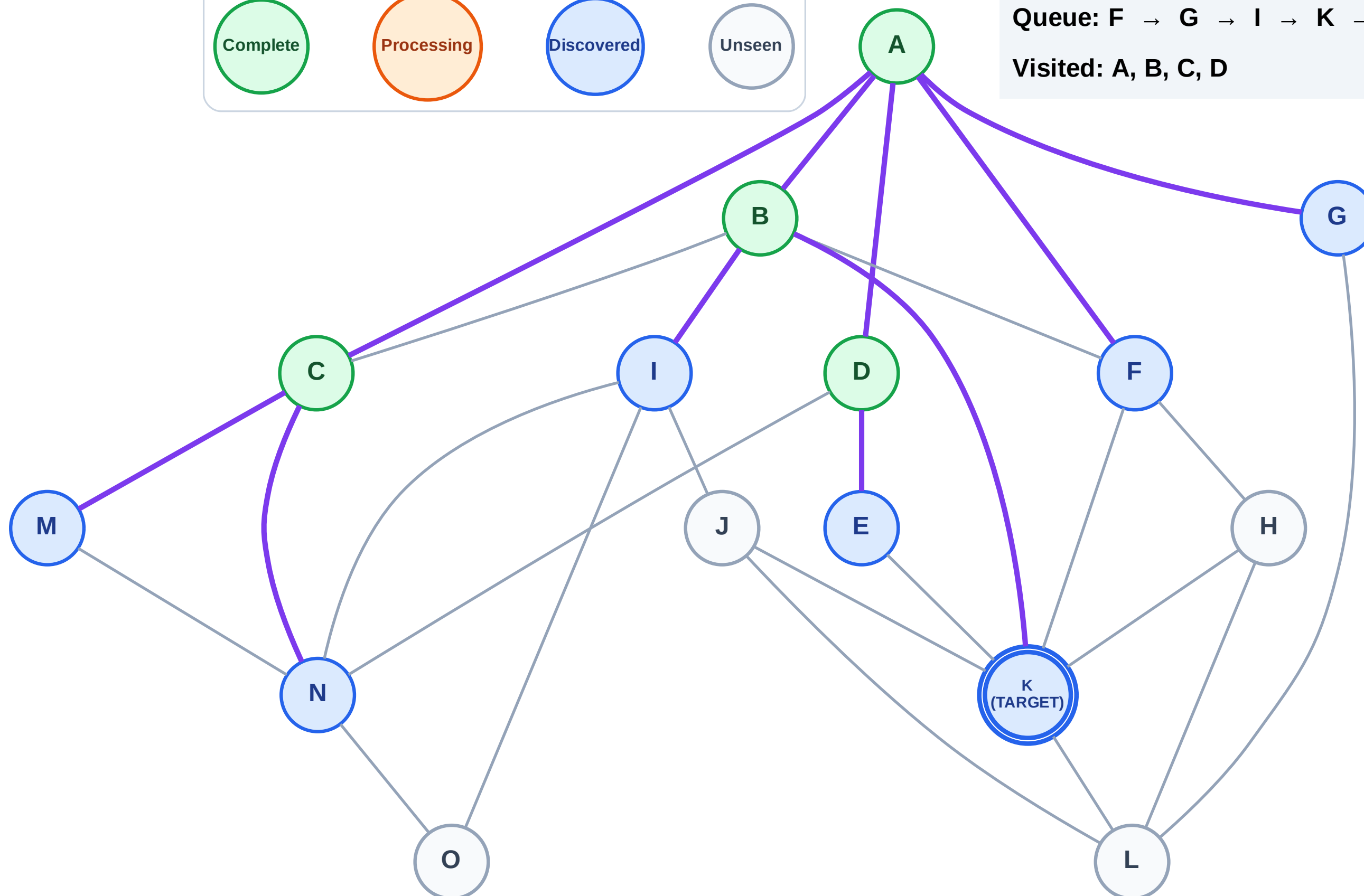
BFS Traversal

Step 9: Discover E from D



Queue: F → G → I → K → M → N → E

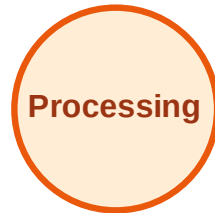
Visited: A, B, C, D



BFS Traversal

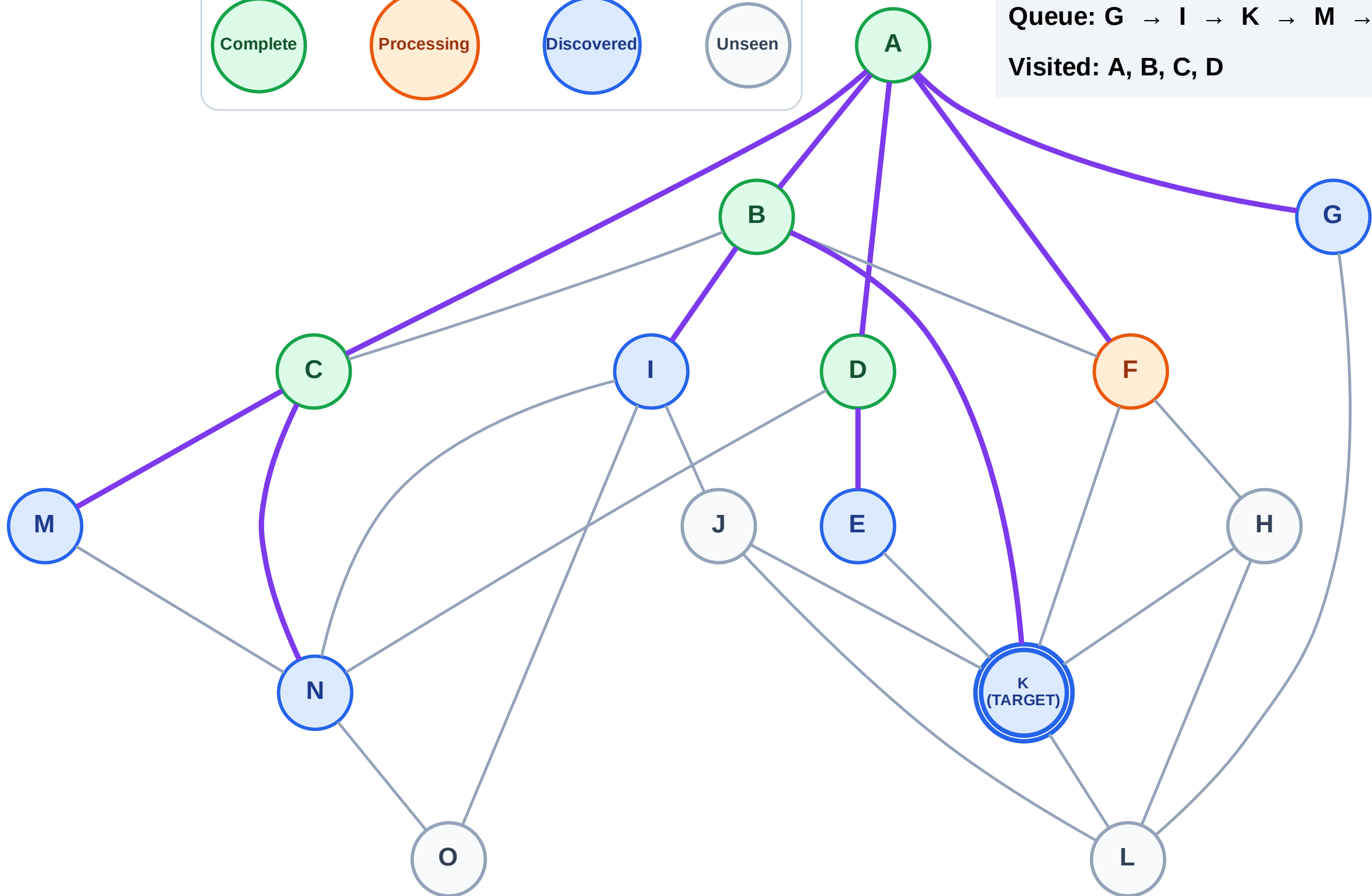
Step 10: Process node F

Legend



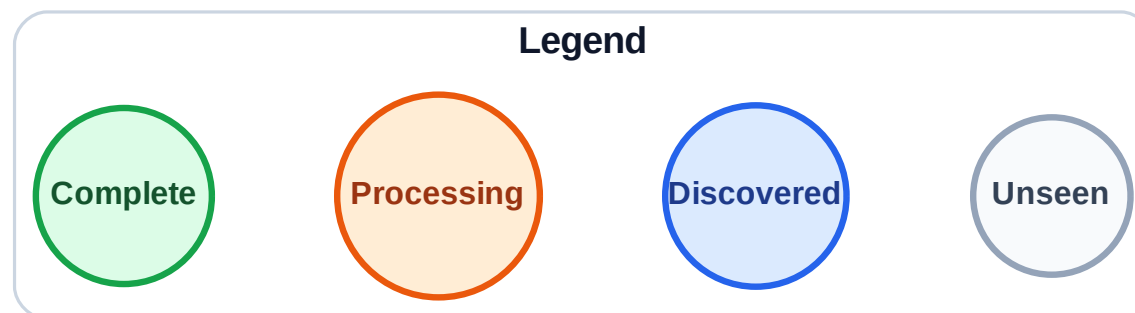
Queue: G → I → K → M → N → E

Visited: A, B, C, D



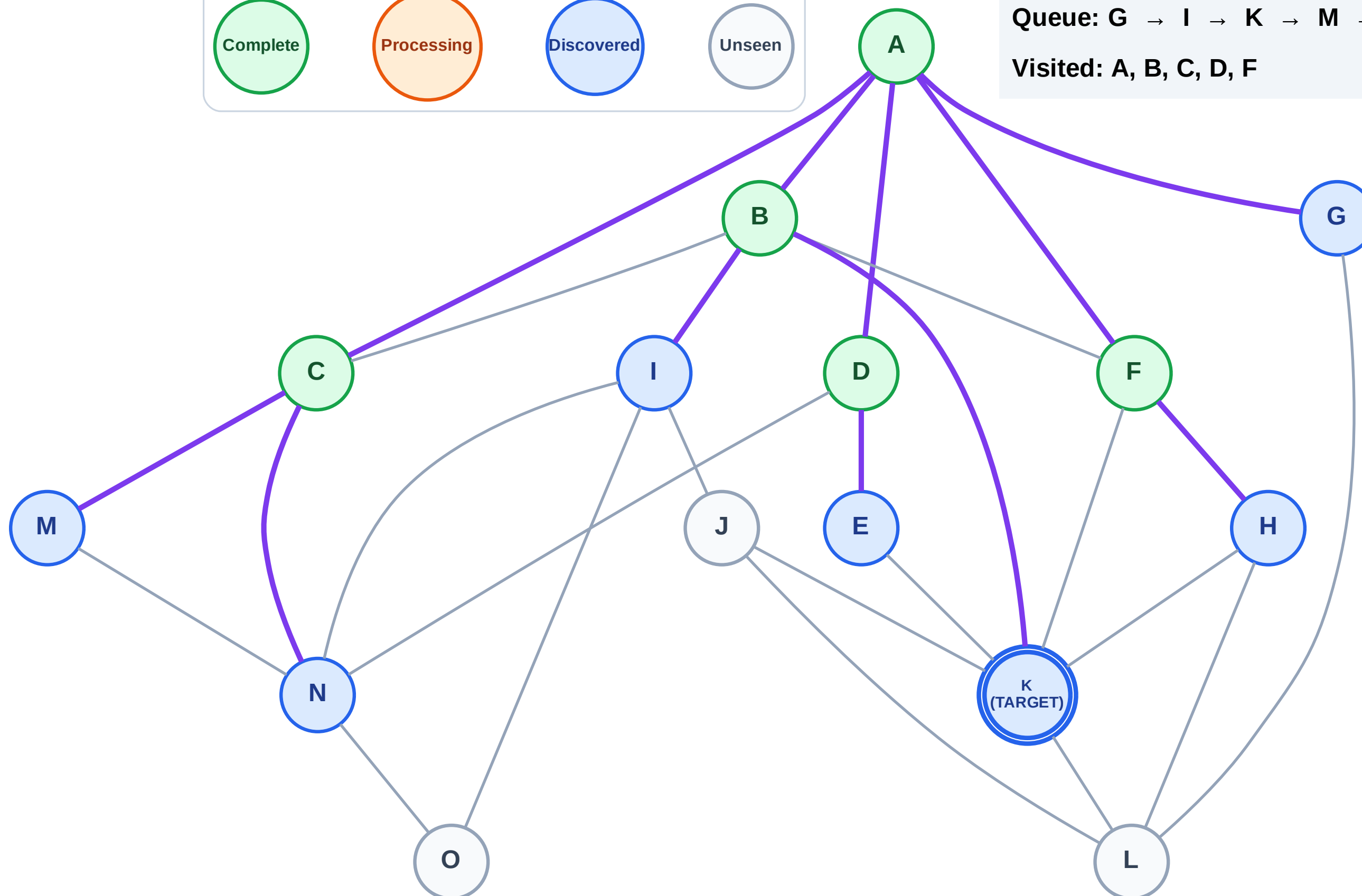
BFS Traversal

Step 11: Discover H from F



Queue: G → I → K → M → N → E → H

Visited: A, B, C, D, F



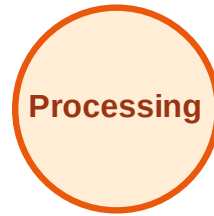
BFS Traversal

Step 12: Process node G

Legend



Complete



Processing



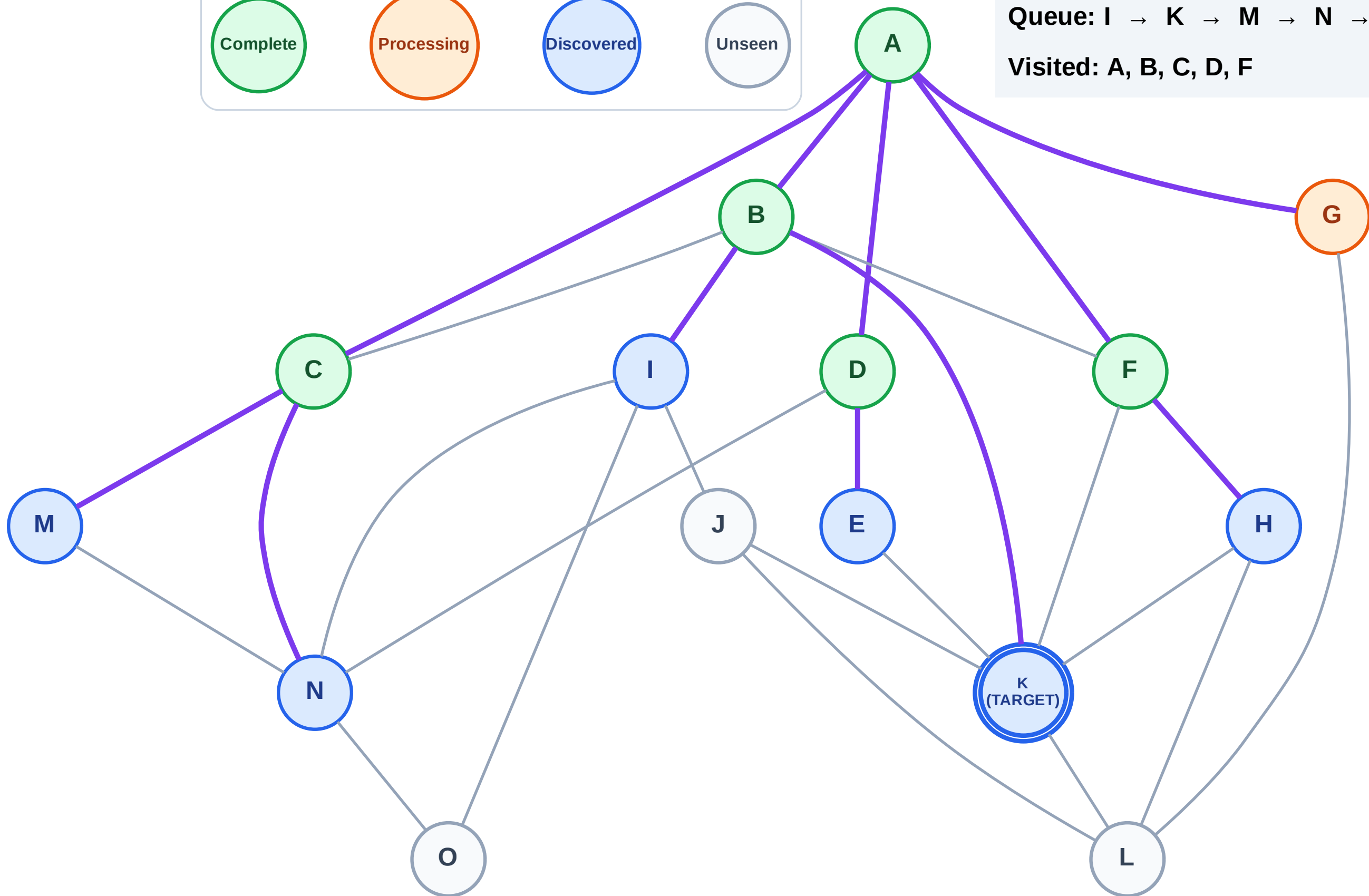
Discovered



Unseen

Queue: I → K → M → N → E → H

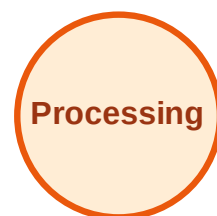
Visited: A, B, C, D, F



BFS Traversal

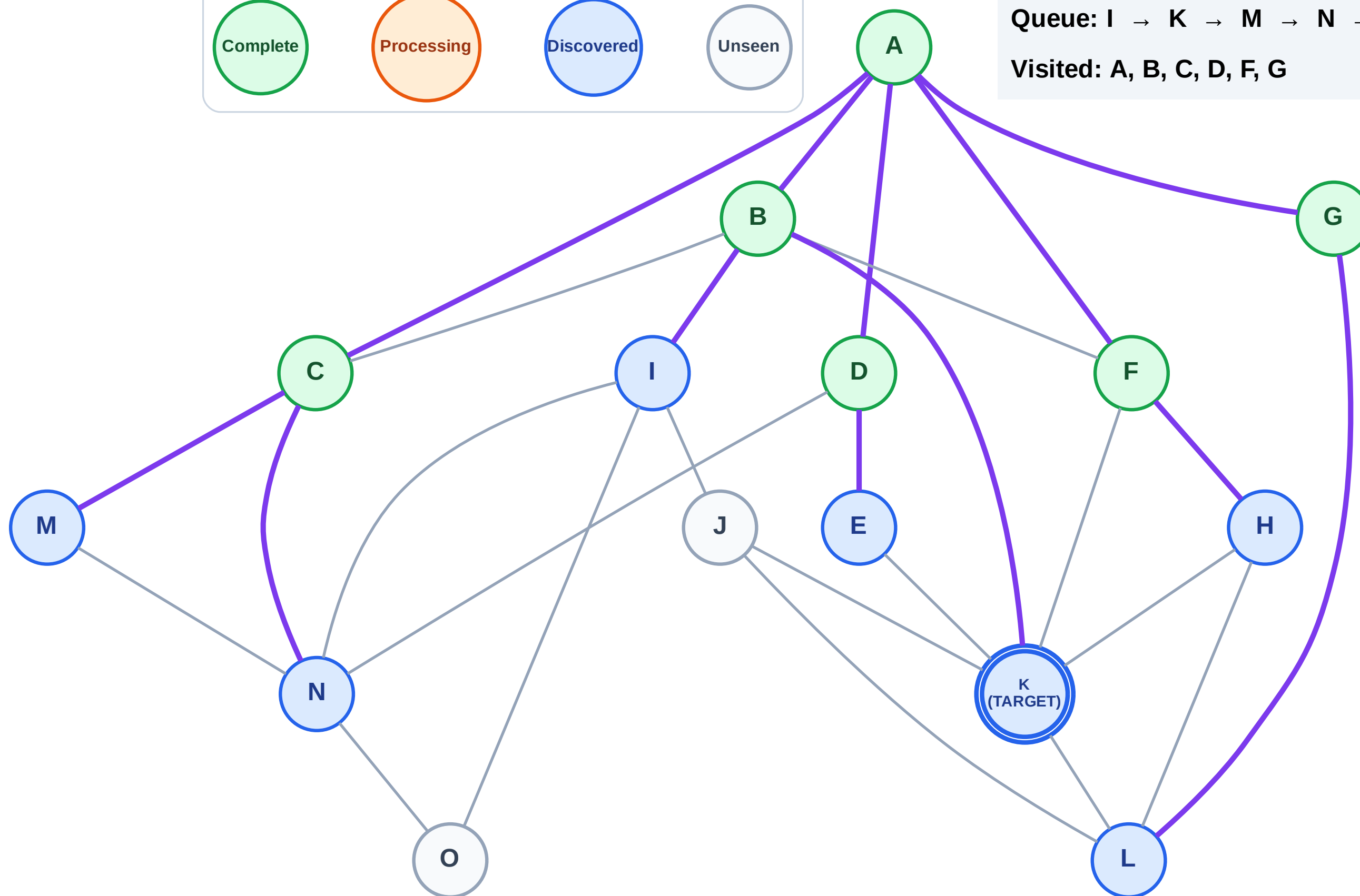
Step 13: Discover L from G

Legend



Queue: I → K → M → N → E → H → L

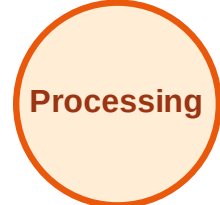
Visited: A, B, C, D, F, G



BFS Traversal

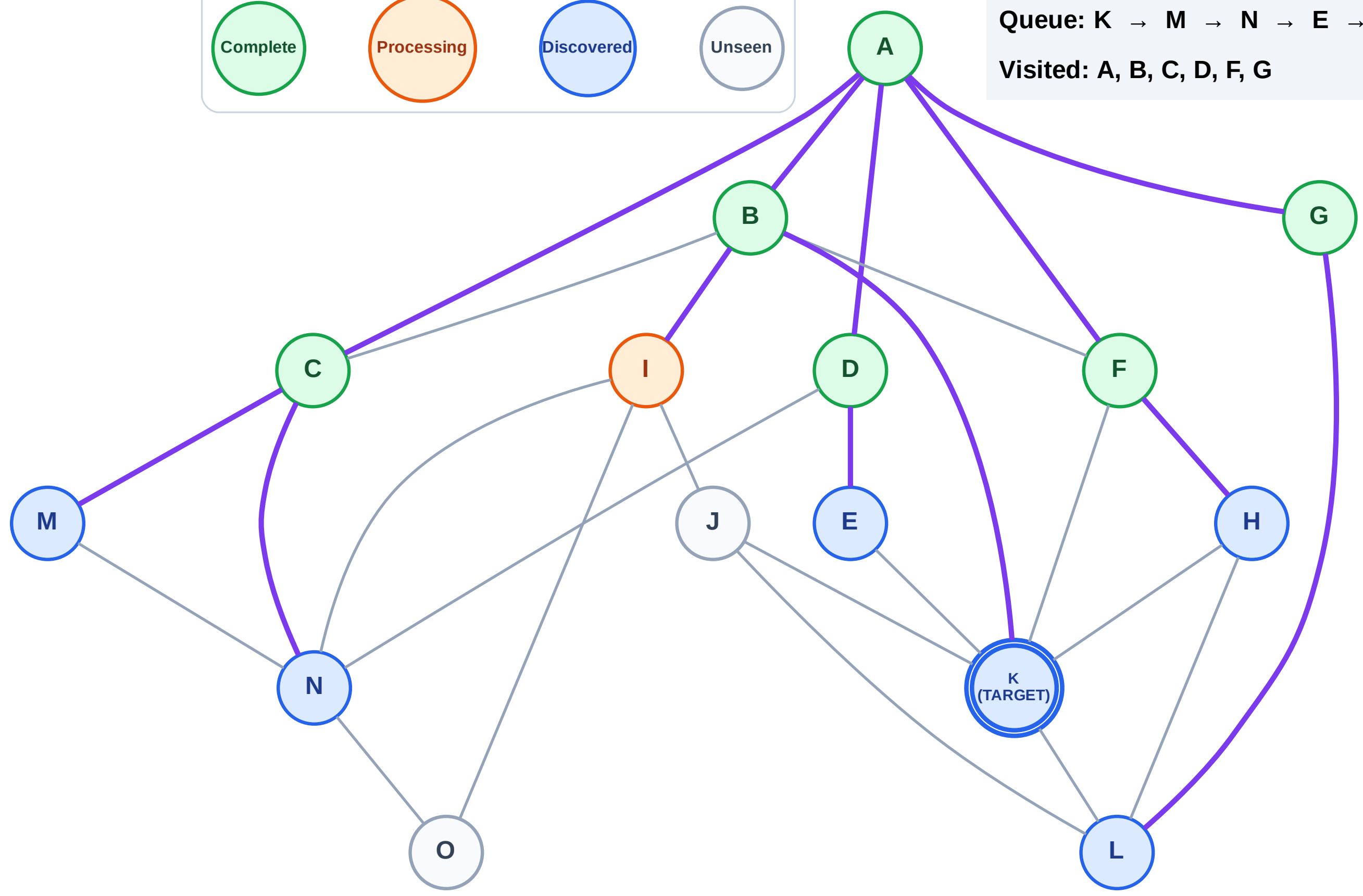
Step 14: Process node I

Legend



Queue: K → M → N → E → H → L

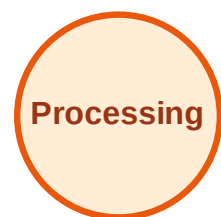
Visited: A, B, C, D, F, G



BFS Traversal

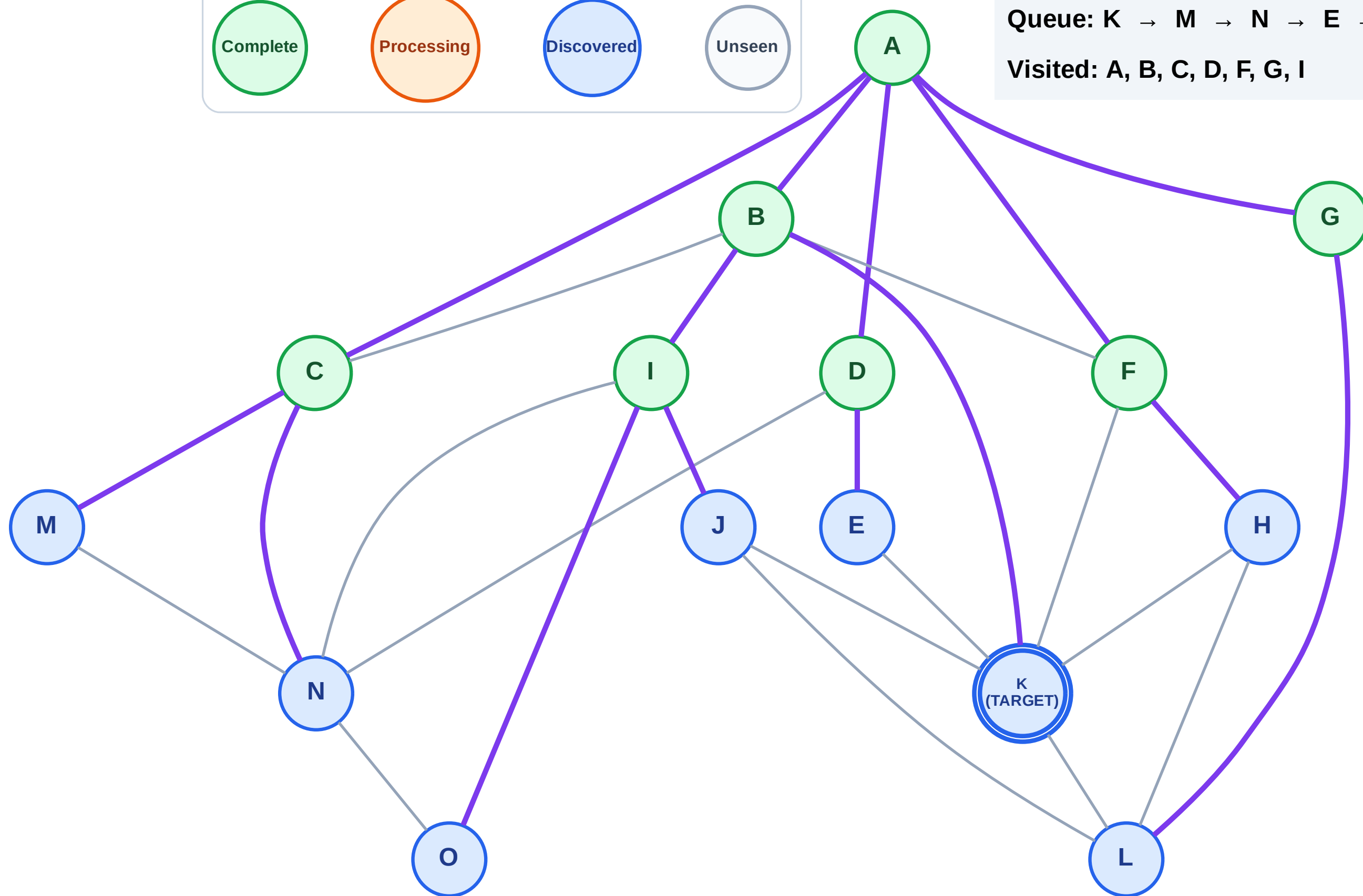
Step 15: Discover J, O from I

Legend



Queue: K → M → N → E → H → L → J → O

Visited: A, B, C, D, F, G, I



BFS Traversal

Step 16: Process node K

Legend

Complete

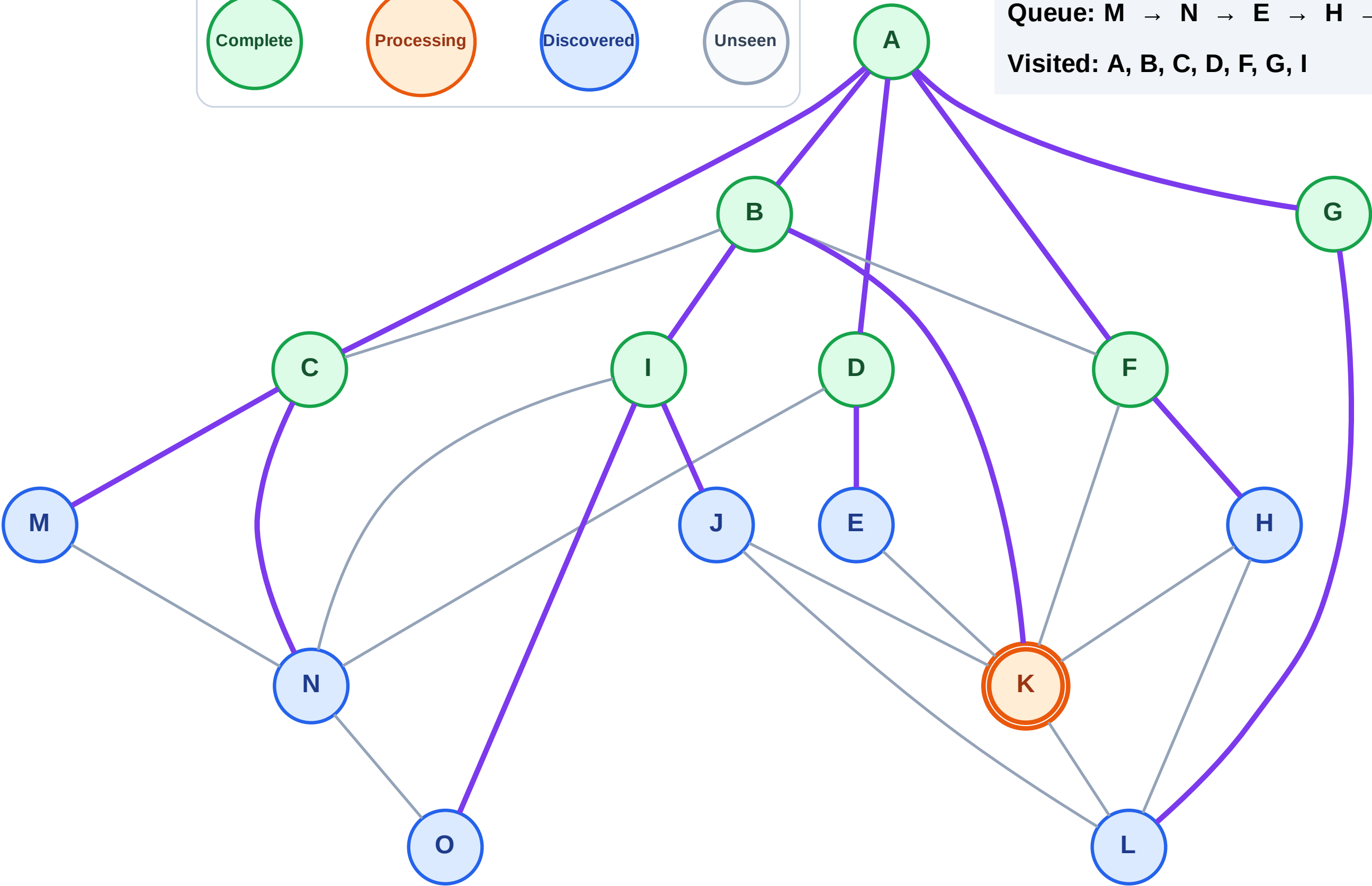
Processing

Discovered

Unseen

Queue: M → N → E → H → L → J → O

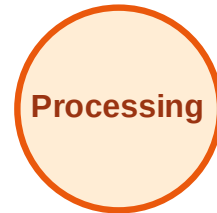
Visited: A, B, C, D, F, G, I



BFS Traversal

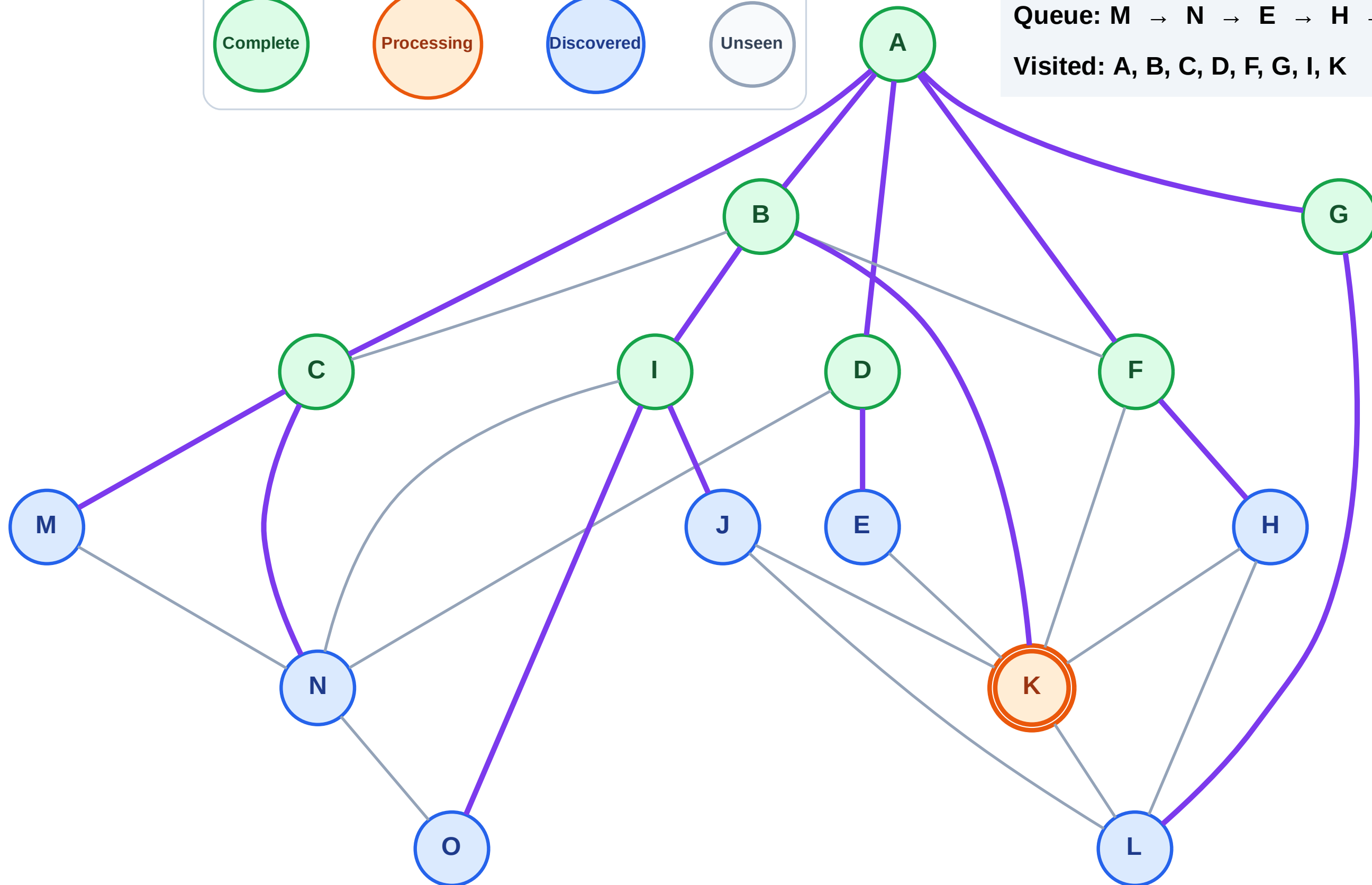
Step 17: TARGET NODE K REACHED! Path (2 hops): A → B → K

Legend



Queue: M → N → E → H → L → J → O

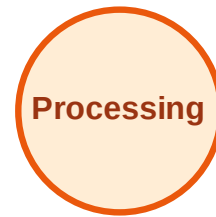
Visited: A, B, C, D, F, G, I, K



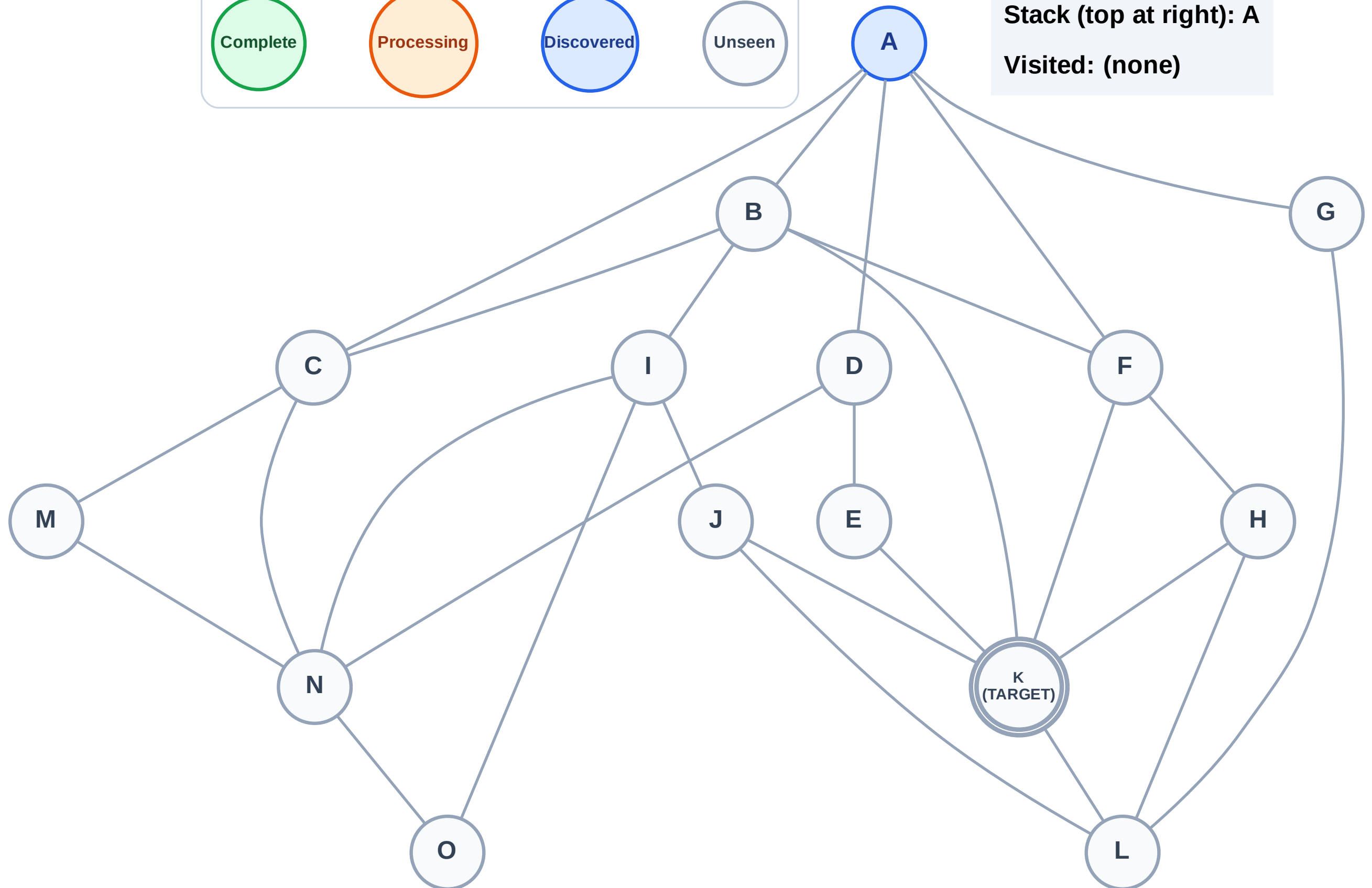
DFS Traversal

Step 1: Start at A (Seeking target node K)

Legend



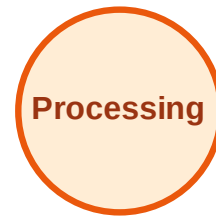
Stack (top at right): A
Visited: (none)



DFS Traversal

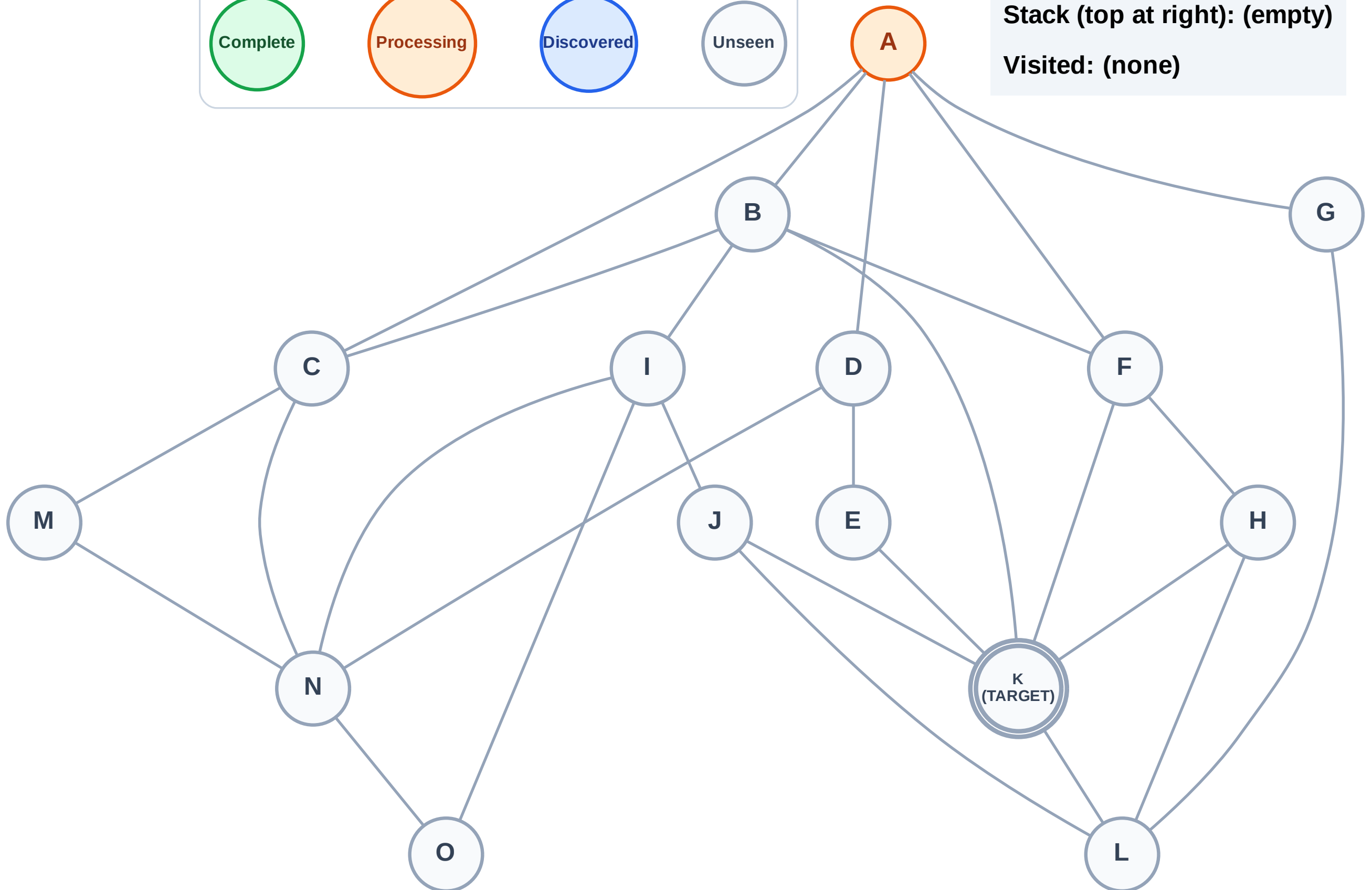
Step 2: Process node A

Legend



Stack (top at right): (empty)

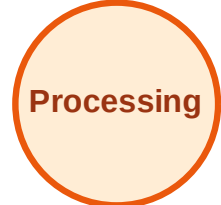
Visited: (none)



DFS Traversal

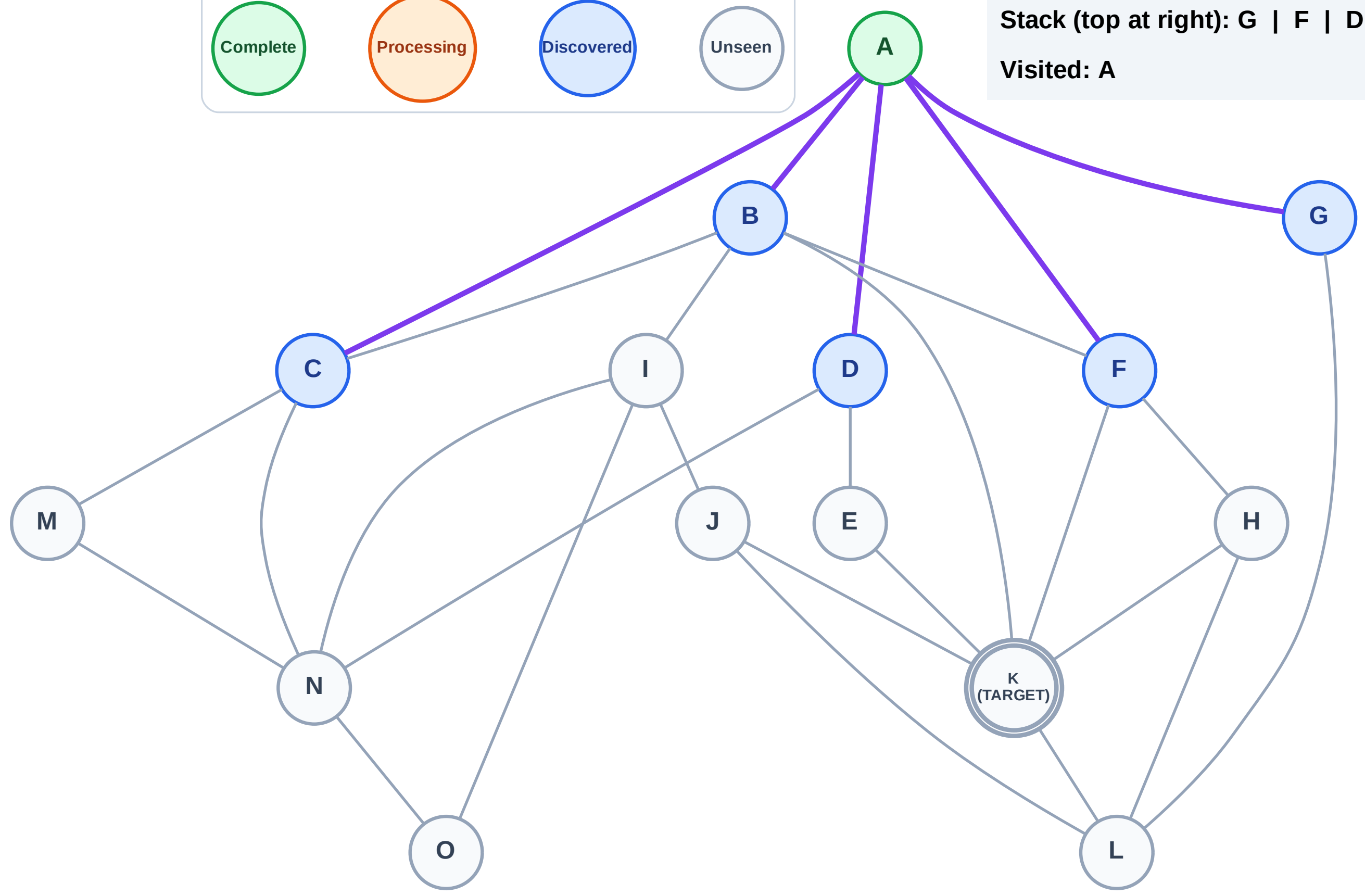
Step 3: Discover G, F, D, C, B from A

Legend



Stack (top at right): G | F | D | C | B

Visited: A



DFS Traversal

Step 4: Process node B

Legend

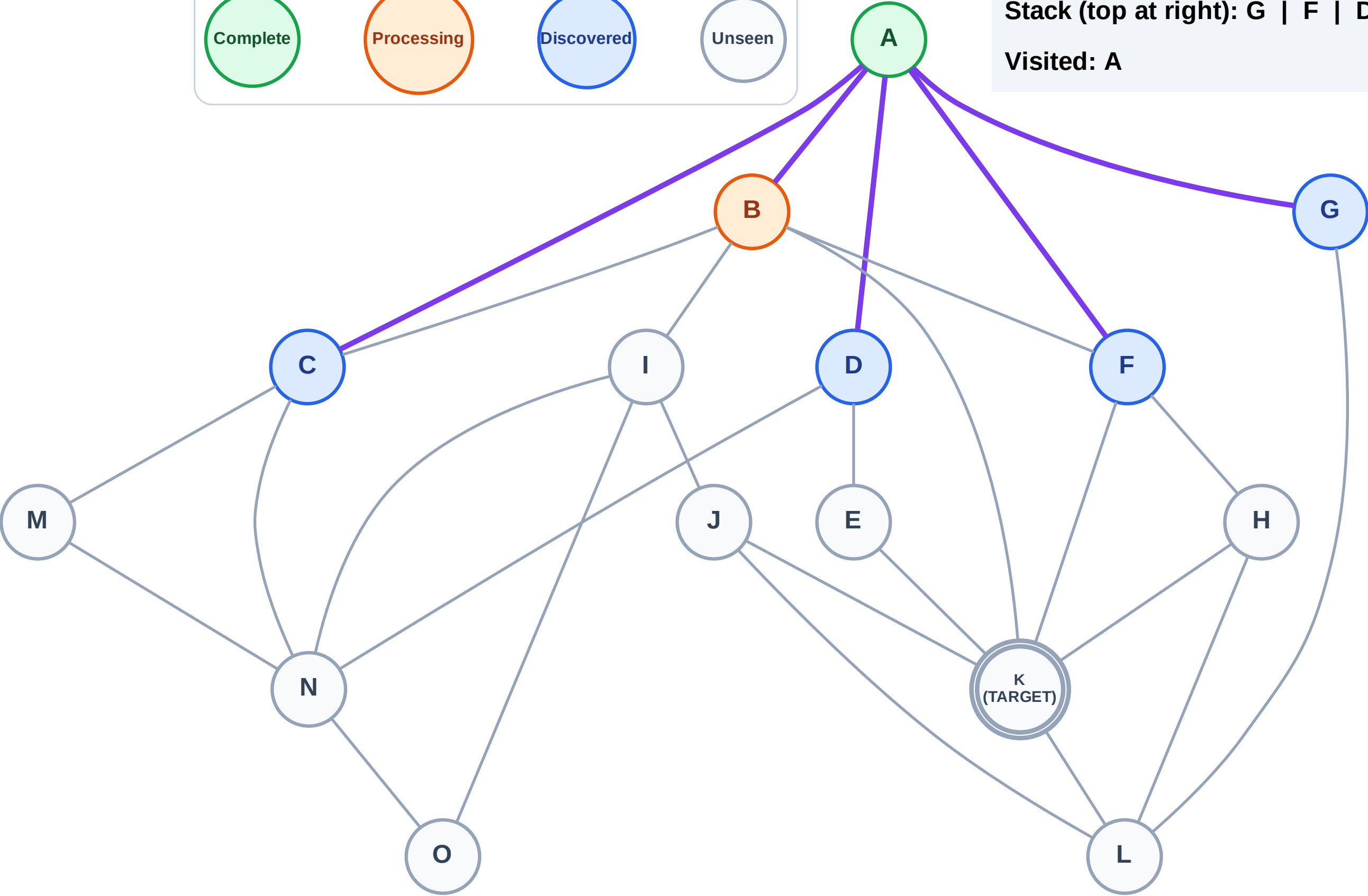
Complete

Processing

Discovered

Unseen

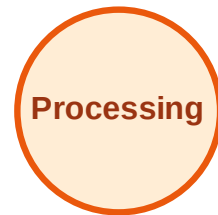
Stack (top at right): G | F | D | C
Visited: A



DFS Traversal

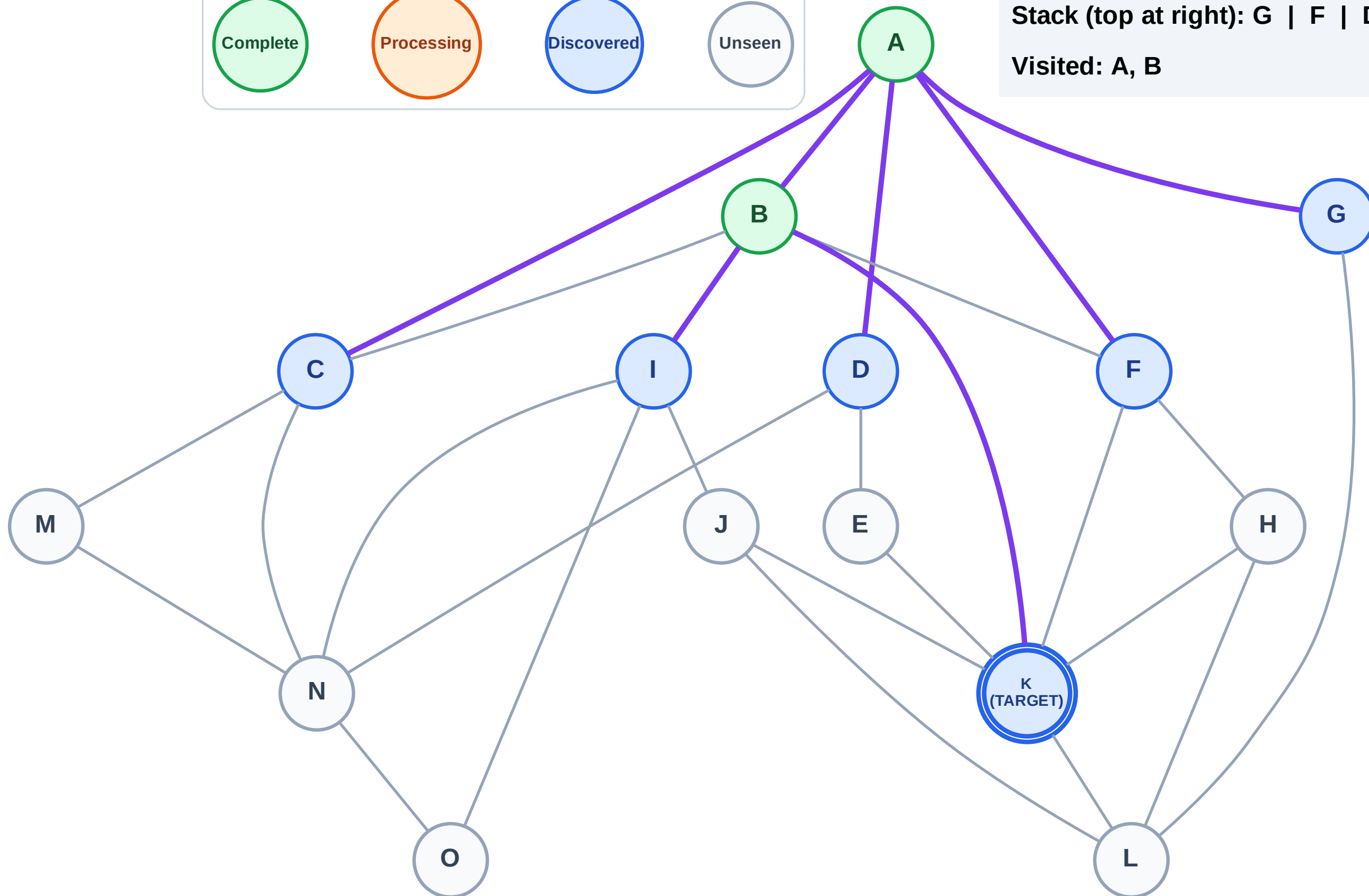
Step 5: Discover K, I from B

Legend



Stack (top at right): G | F | D | C | K | I

Visited: A, B



DFS Traversal

Step 6: Process node I

Legend

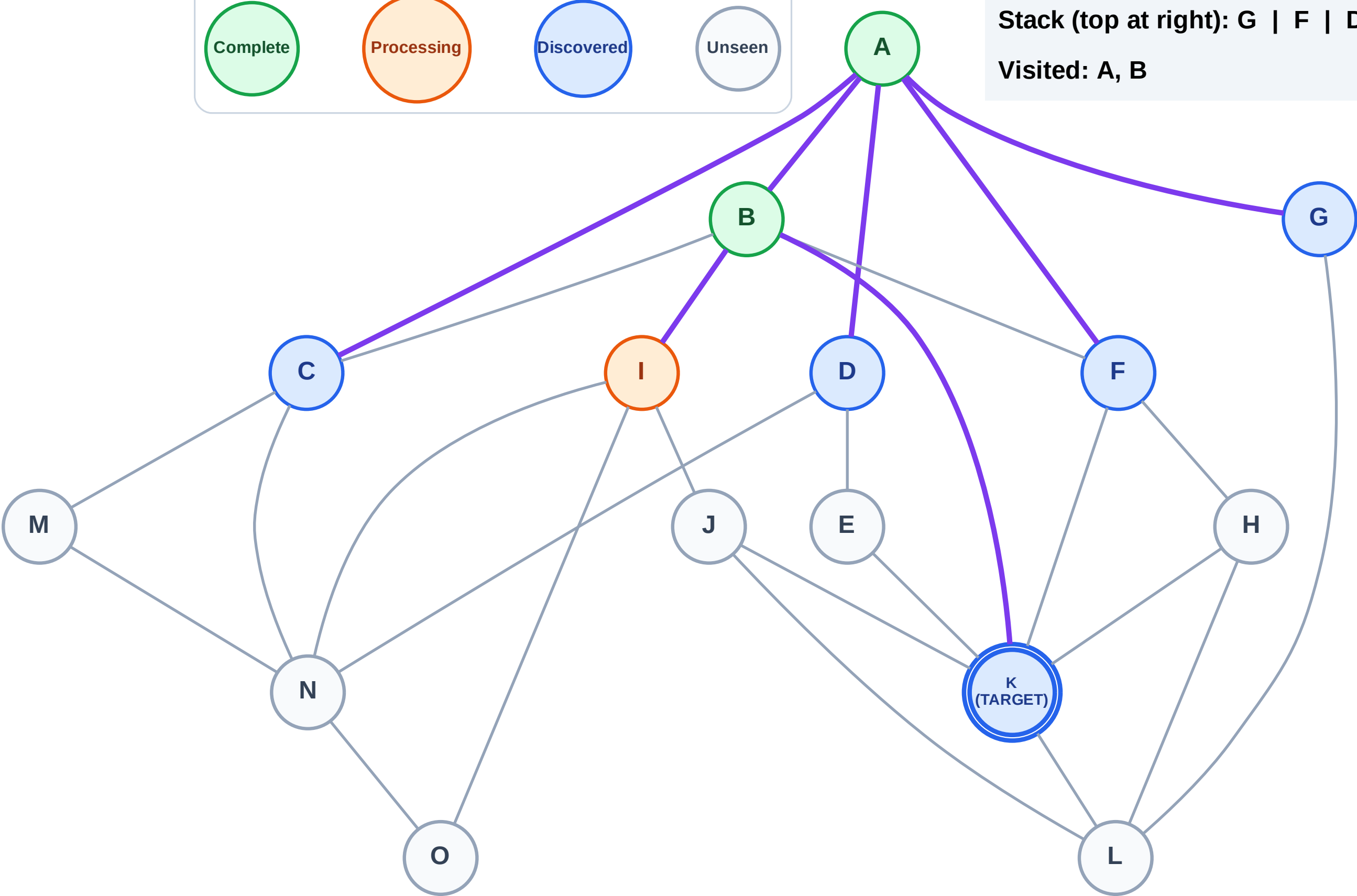
Complete

Processing

Discovered

Unseen

Stack (top at right): G | F | D | C | K
Visited: A, B



DFS Traversal

Step 7: Discover O, N, J from I

Legend

Complete

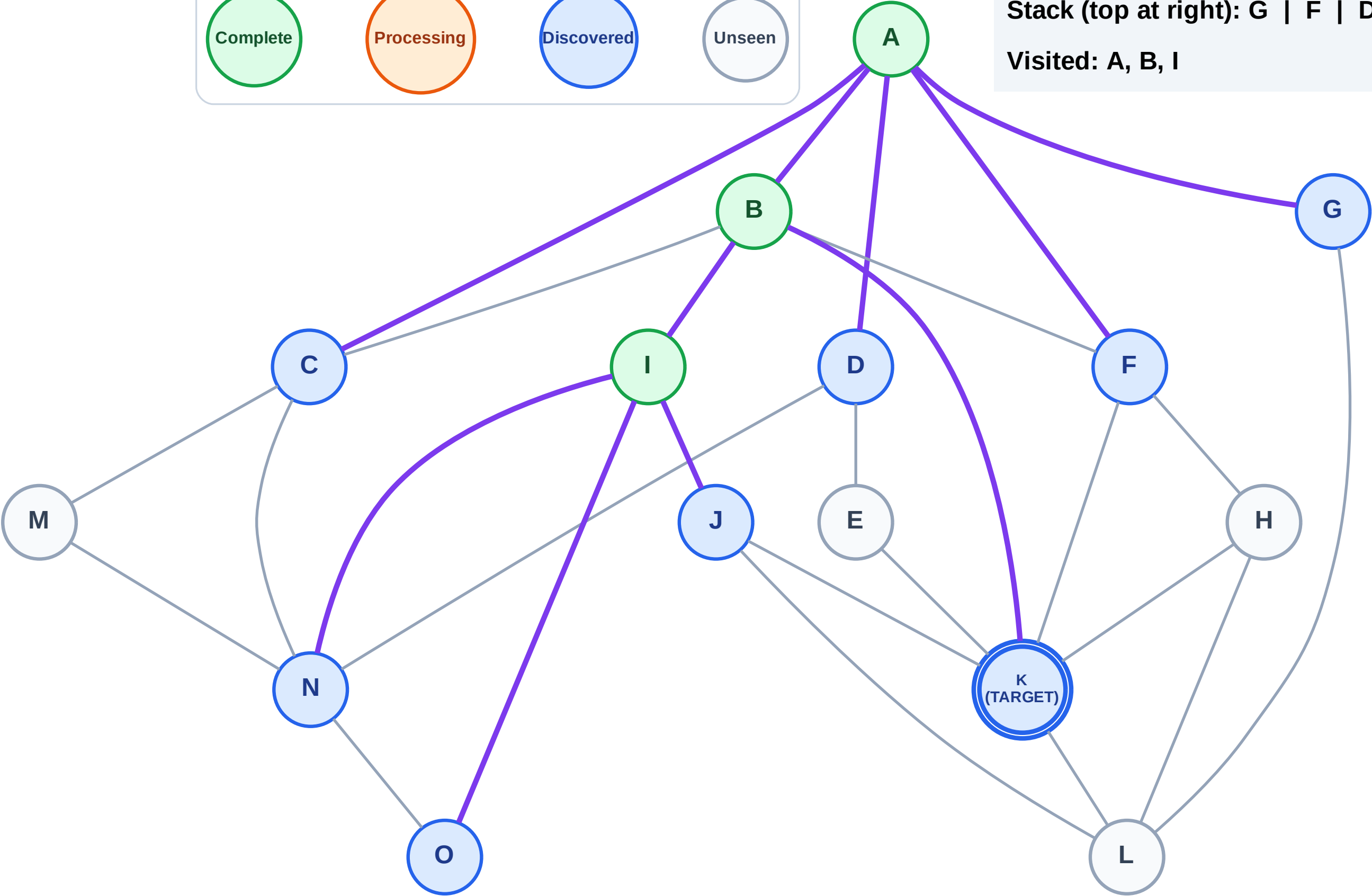
Processing

Discovered

Unseen

Stack (top at right): G | F | D | C | K | O | N | J

Visited: A, B, I



DFS Traversal

Step 8: Process node J

Legend

Complete

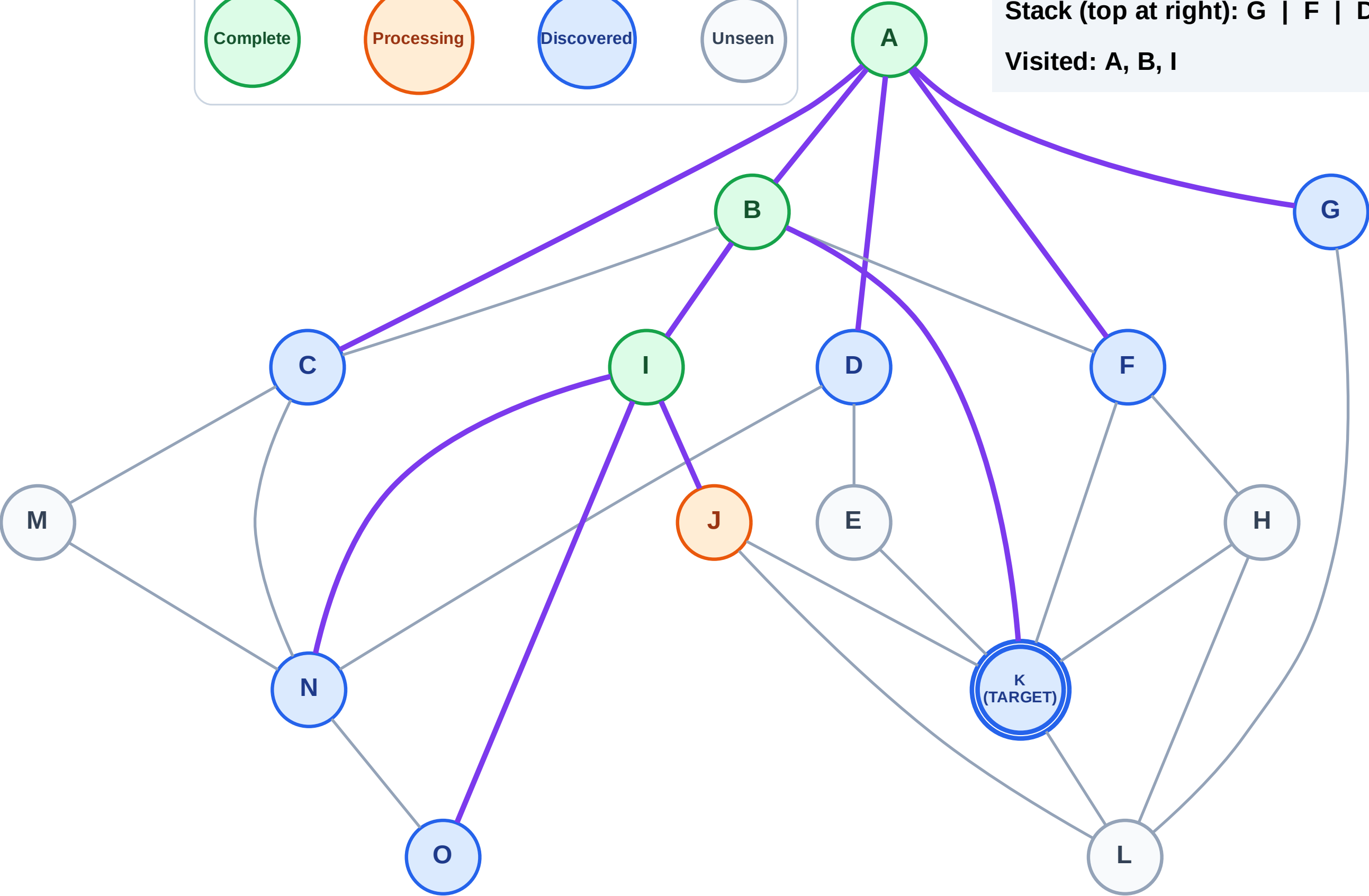
Processing

Discovered

Unseen

Stack (top at right): G | F | D | C | K | O | N

Visited: A, B, I



DFS Traversal

Step 9: Discover L from J

Legend

Complete

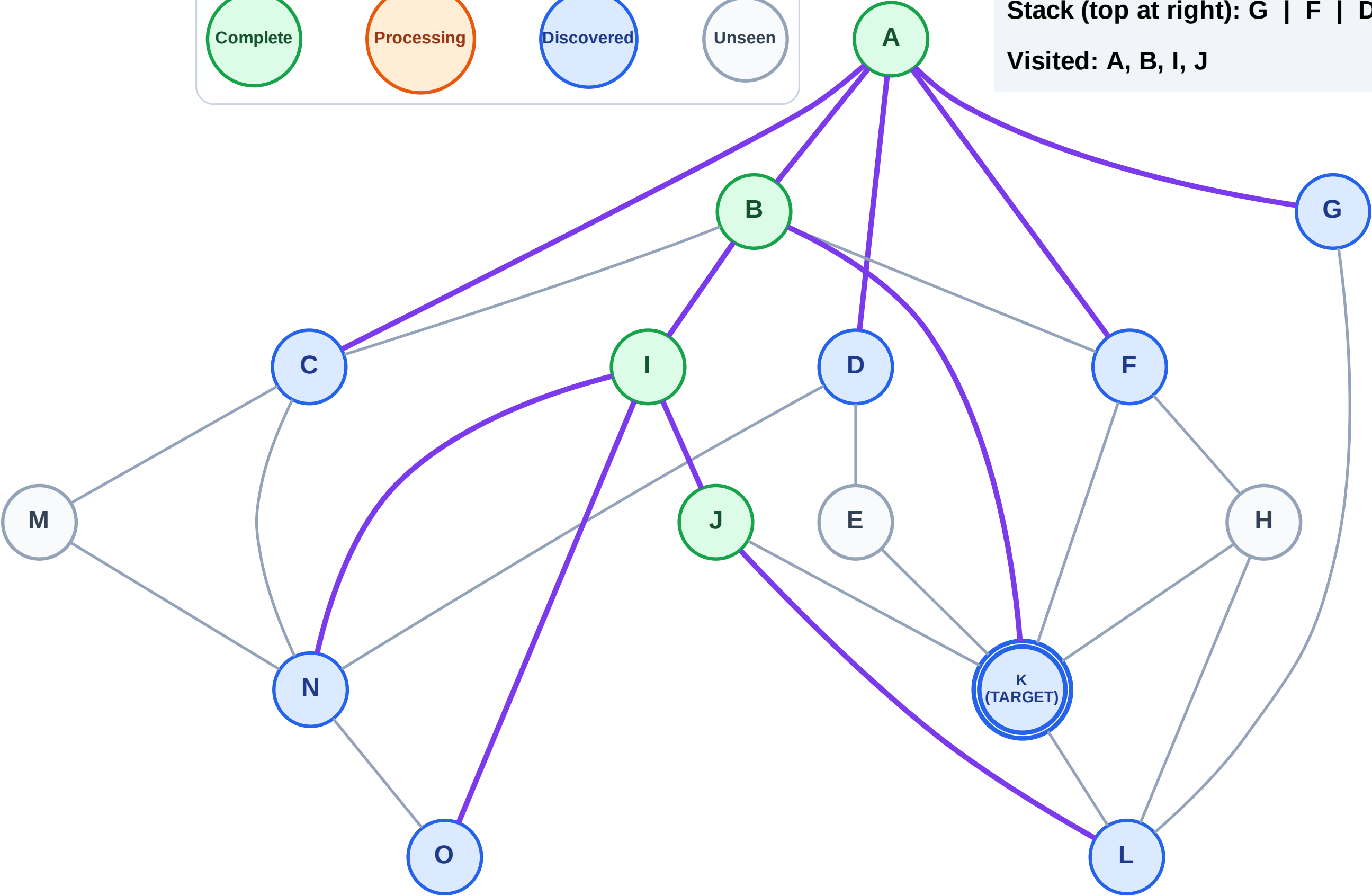
Processing

Discovered

Unseen

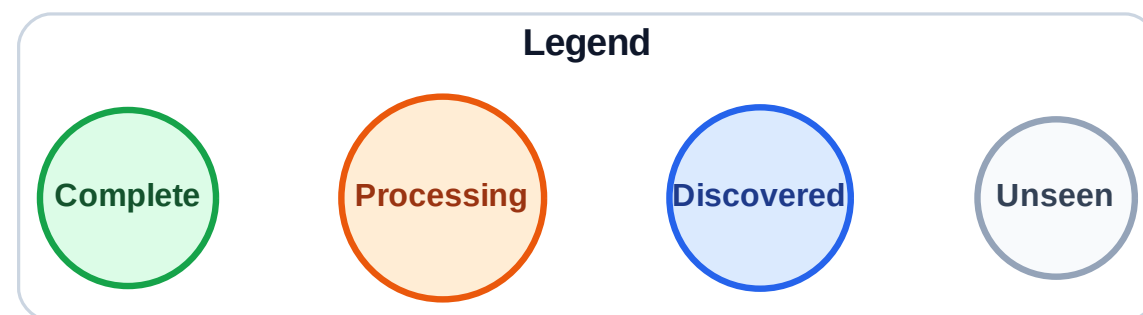
Stack (top at right): G | F | D | C | K | O | N | L

Visited: A, B, I, J

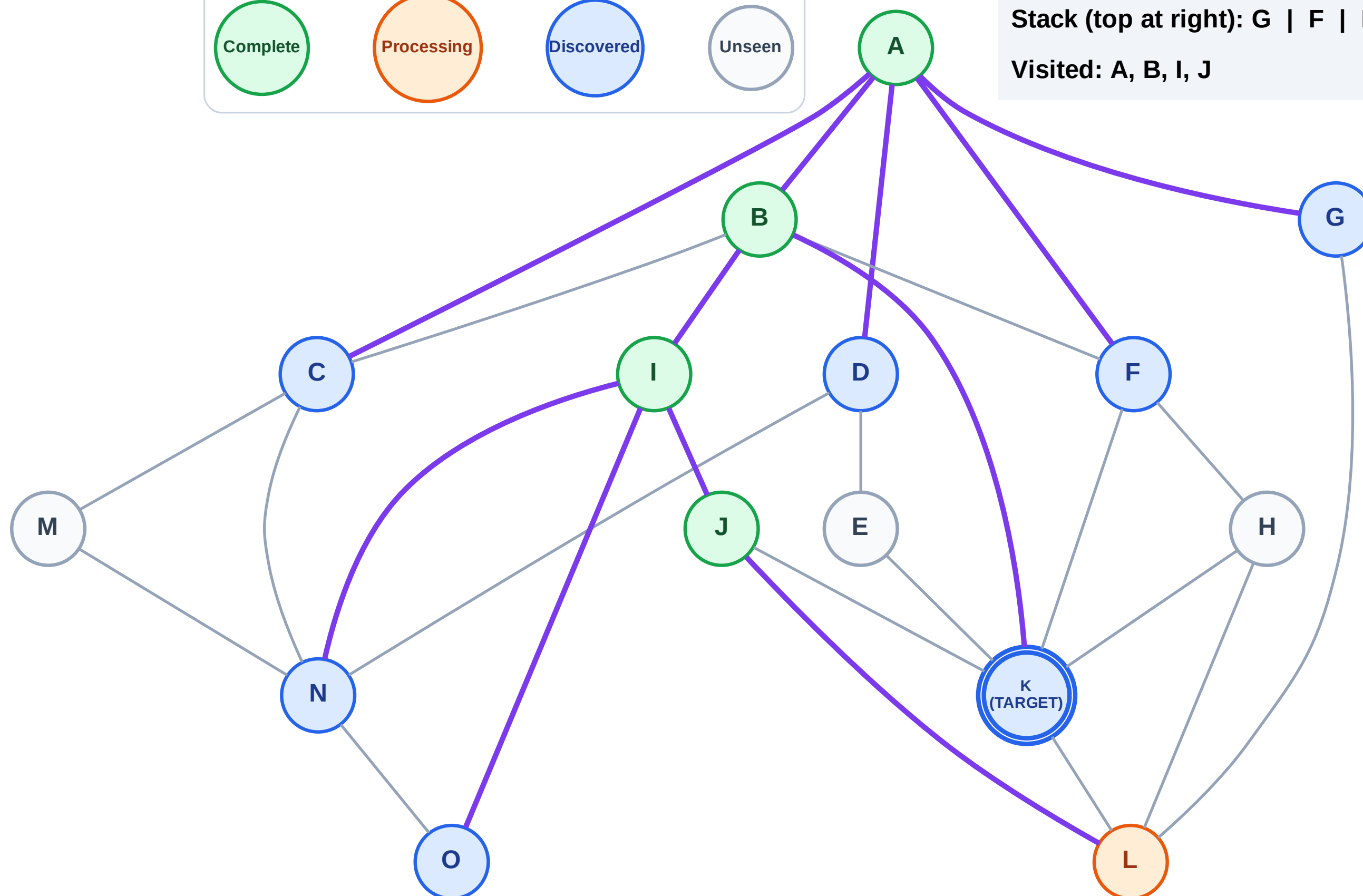


DFS Traversal

Step 10: Process node L



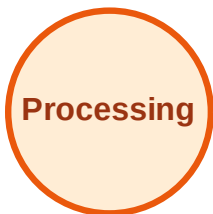
Stack (top at right): G | F | D | C | K | O | N
Visited: A, B, I, J



DFS Traversal

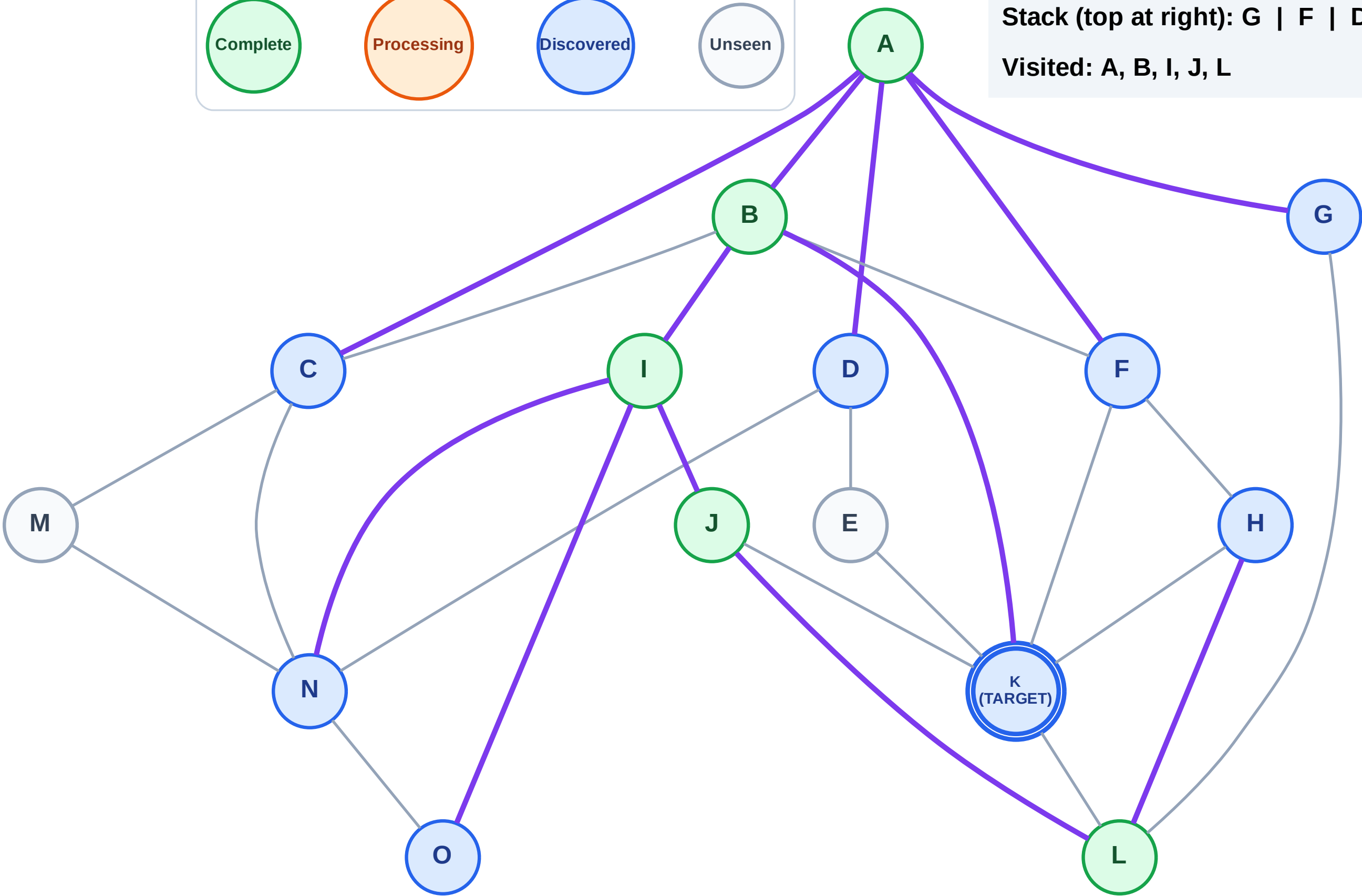
Step 11: Discover H from L

Legend



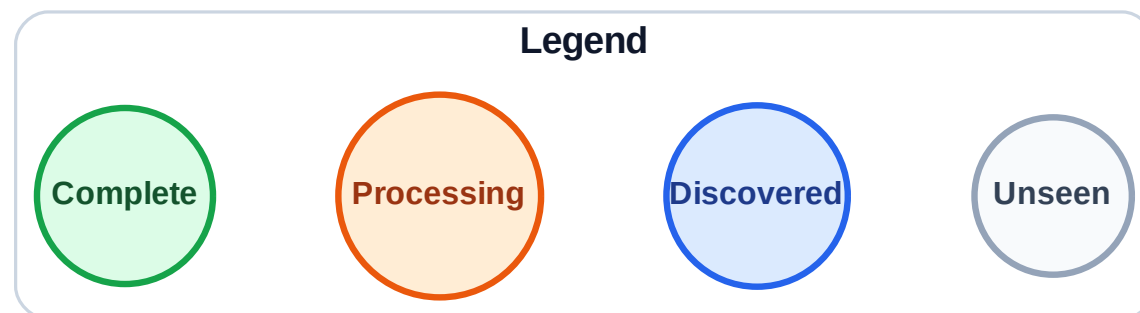
Stack (top at right): G | F | D | C | K | O | N | H

Visited: A, B, I, J, L



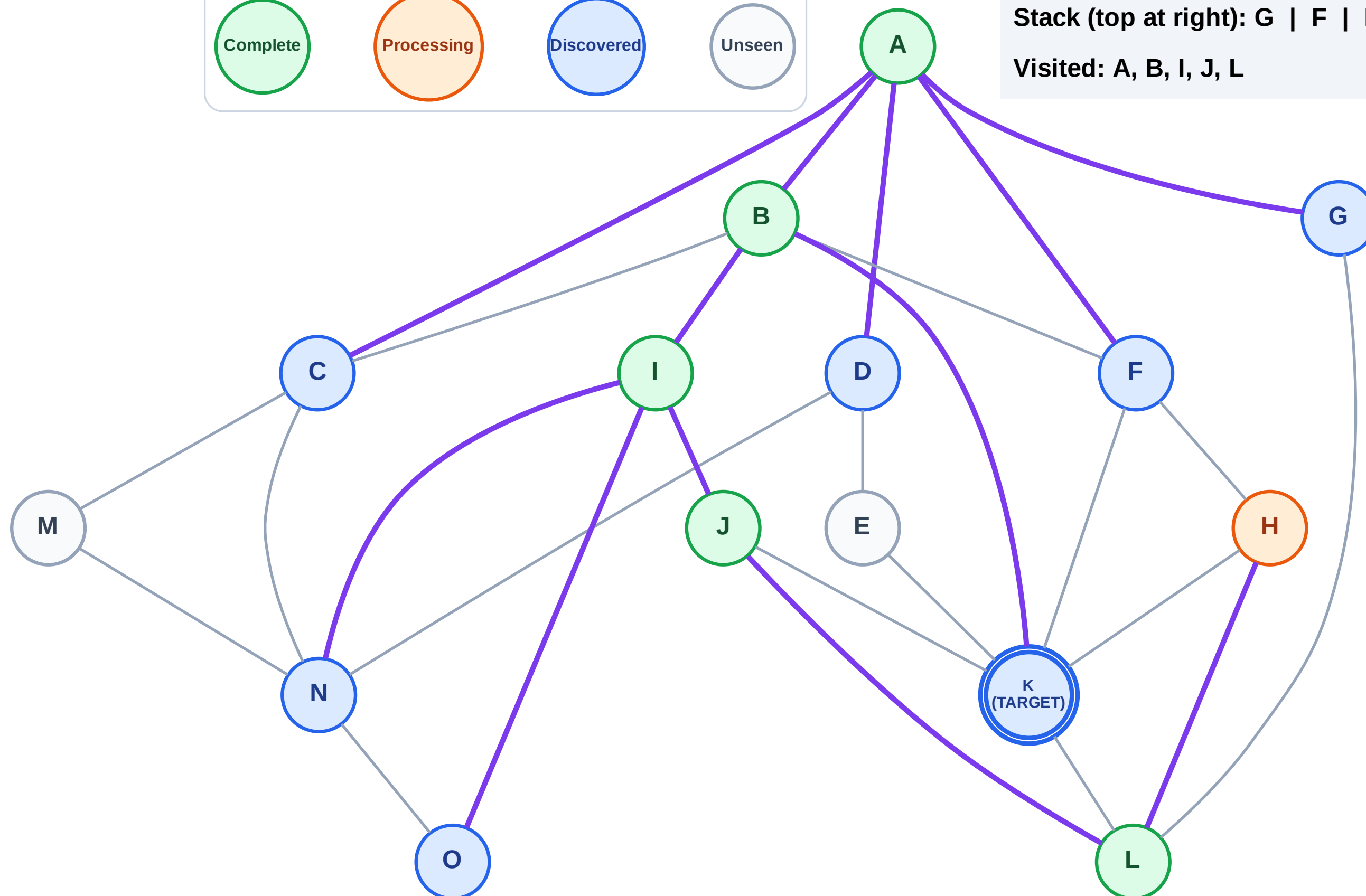
DFS Traversal

Step 12: Process node H



Stack (top at right): G | F | D | C | K | O | N

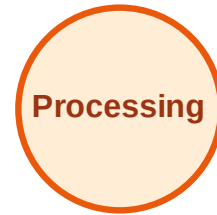
Visited: A, B, I, J, L



DFS Traversal

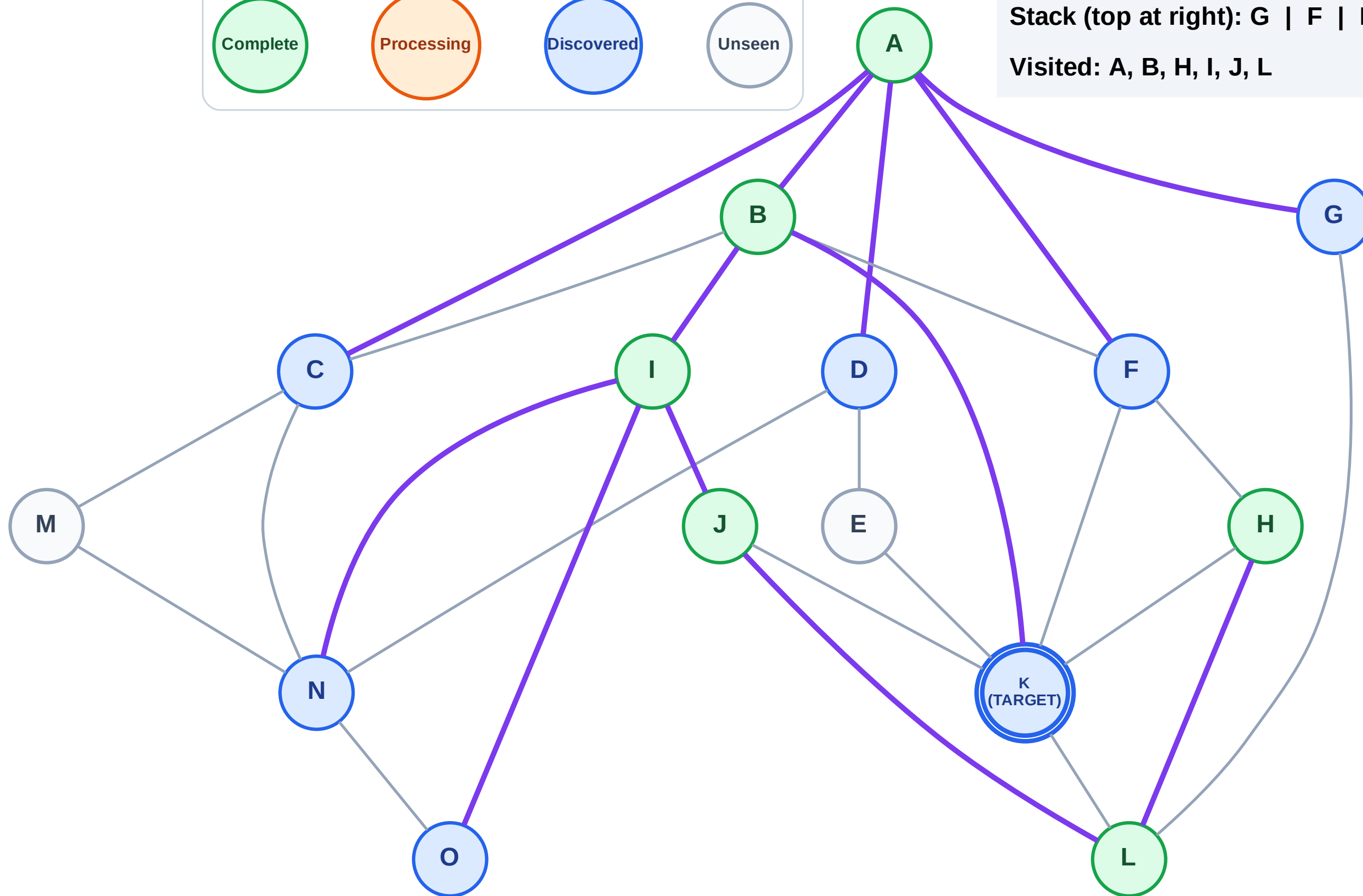
Step 13: No new neighbors from H

Legend



Stack (top at right): G | F | D | C | K | O | N

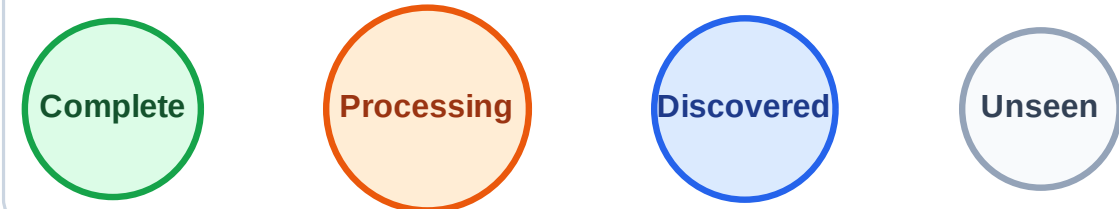
Visited: A, B, H, I, J, L



DFS Traversal

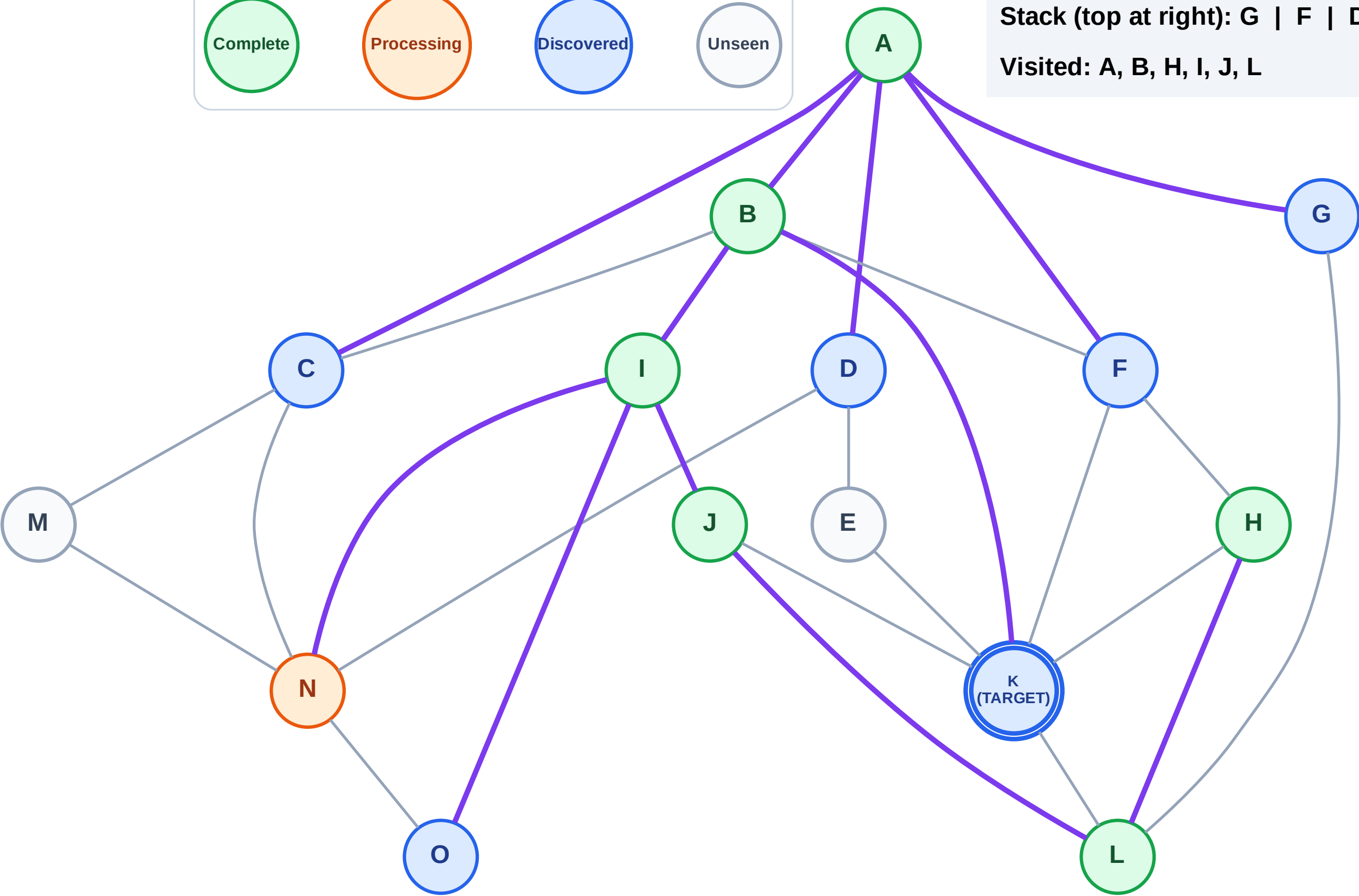
Step 14: Process node N

Legend



Stack (top at right): G | F | D | C | K | O

Visited: A, B, H, I, J, L



DFS Traversal

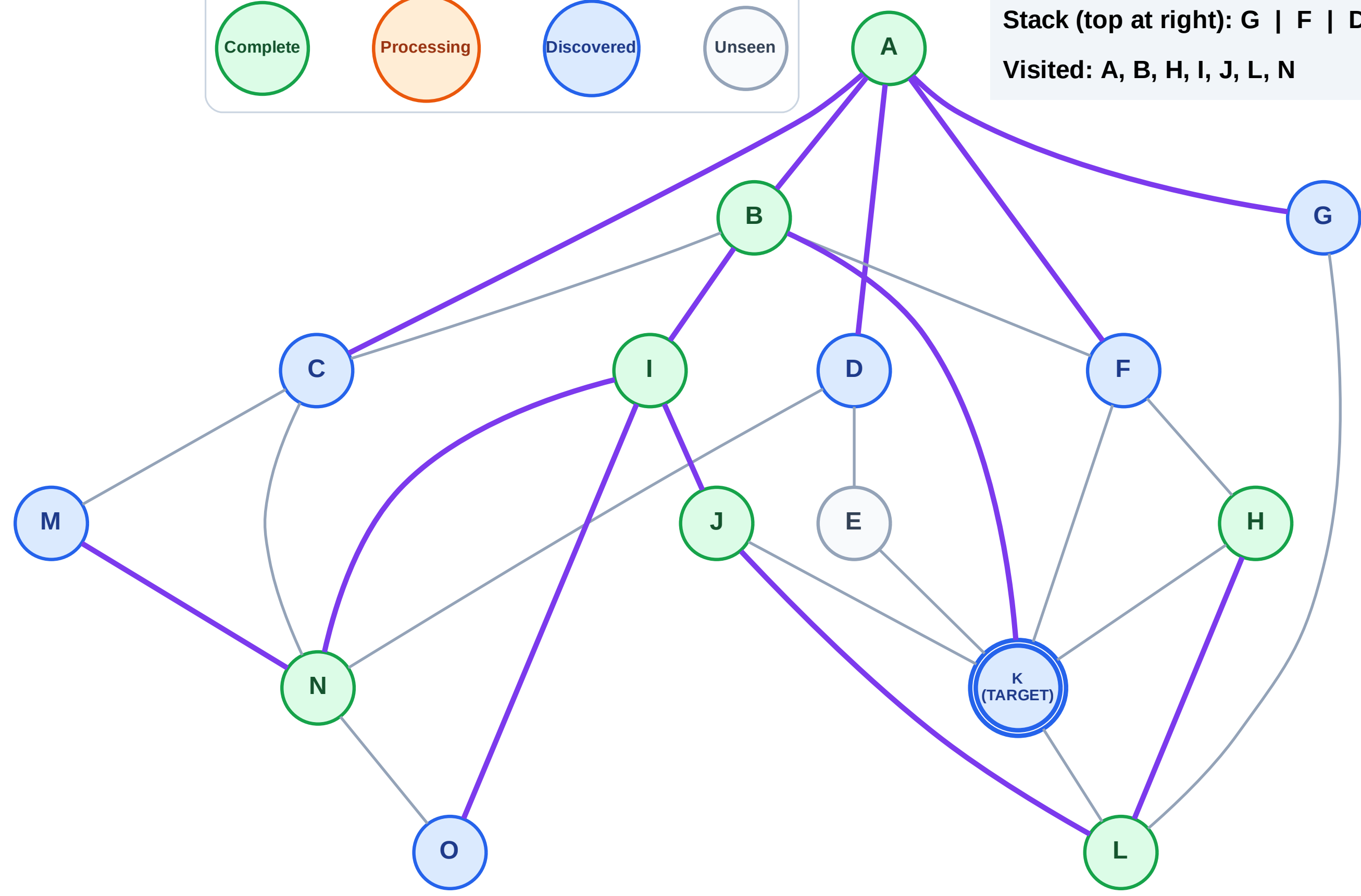
Step 15: Discover M from N

Legend



Stack (top at right): G | F | D | C | K | O | M

Visited: A, B, H, I, J, L, N



DFS Traversal

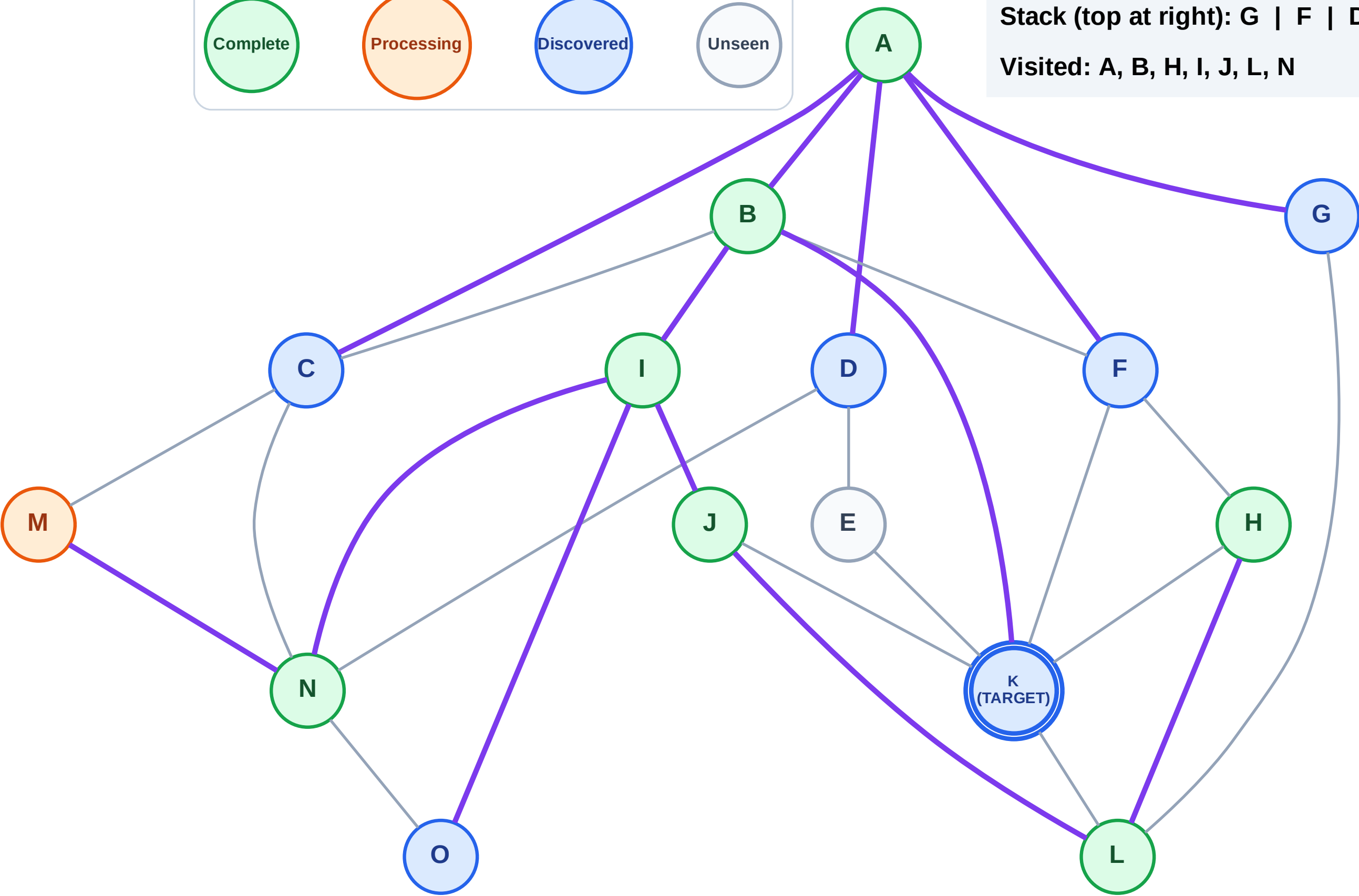
Step 16: Process node M

Legend



Stack (top at right): G | F | D | C | K | O

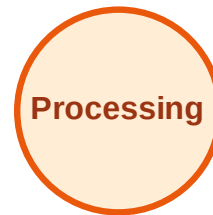
Visited: A, B, H, I, J, L, N



DFS Traversal

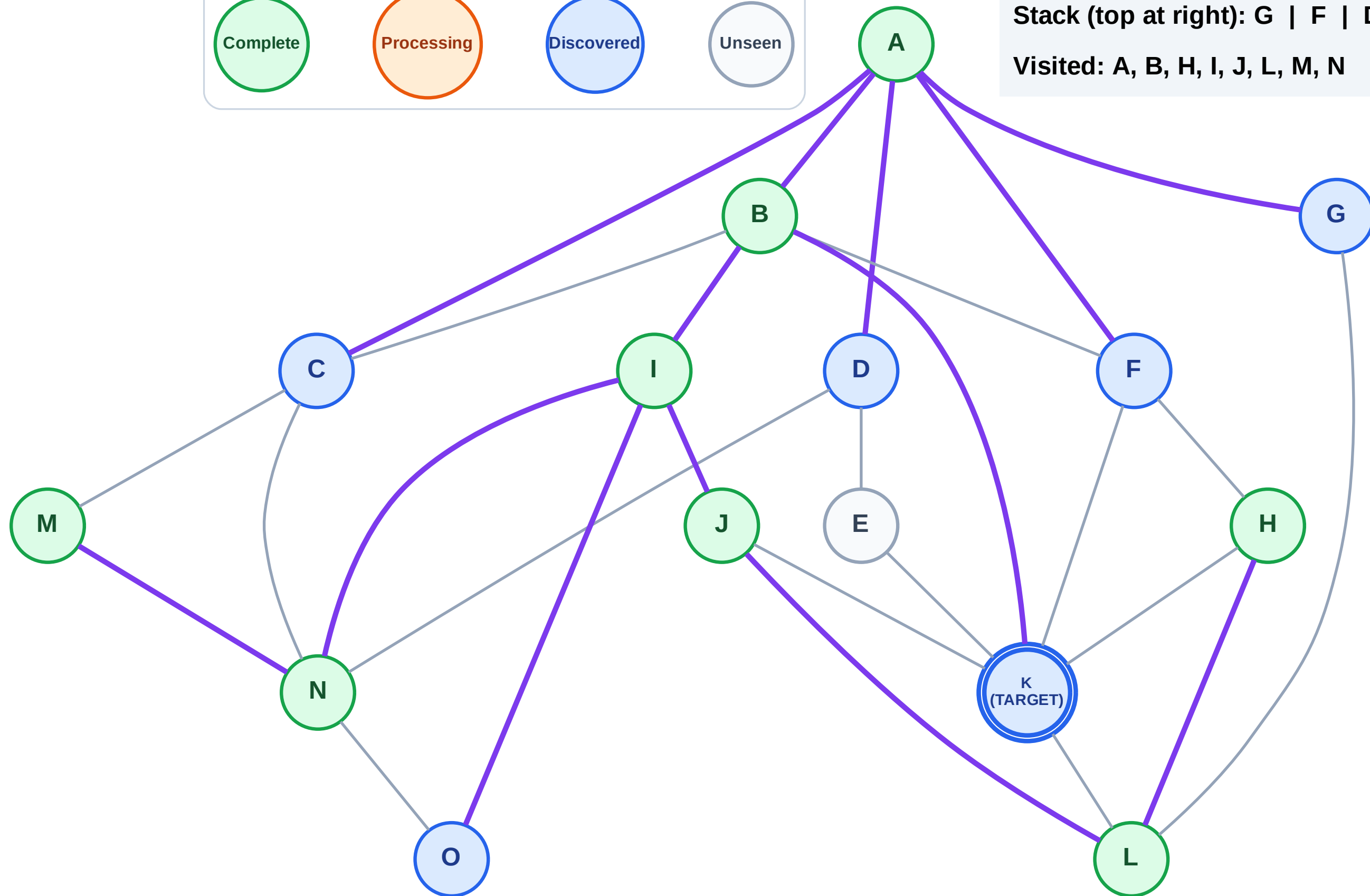
Step 17: No new neighbors from M

Legend



Stack (top at right): G | F | D | C | K | O

Visited: A, B, H, I, J, L, M, N



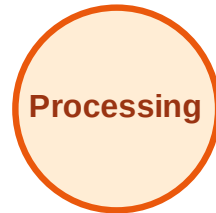
DFS Traversal

Step 18: Process node O

Legend



Complete



Processing



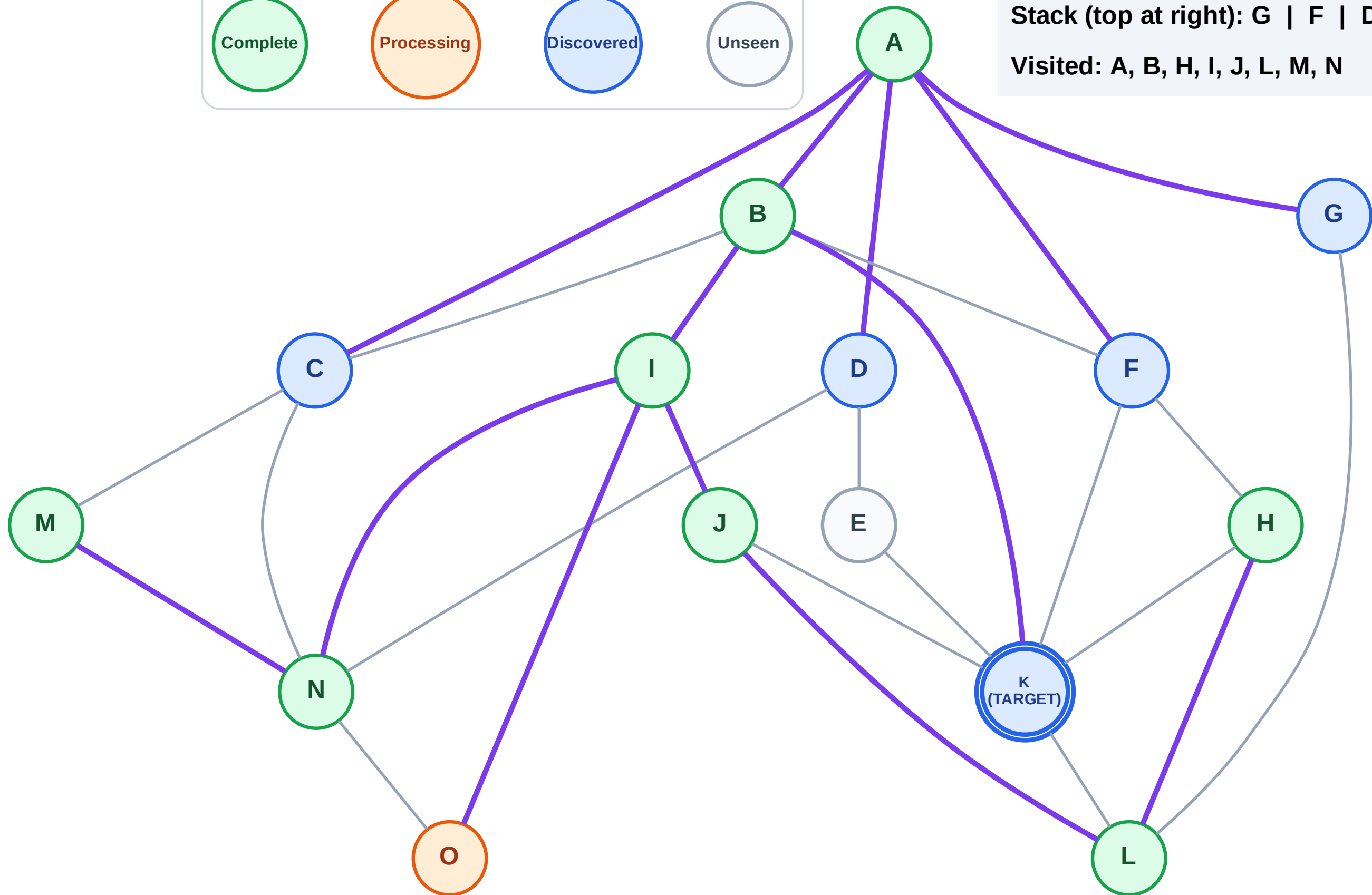
Discovered



Unseen

Stack (top at right): G | F | D | C | K

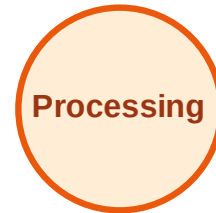
Visited: A, B, H, I, J, L, M, N



DFS Traversal

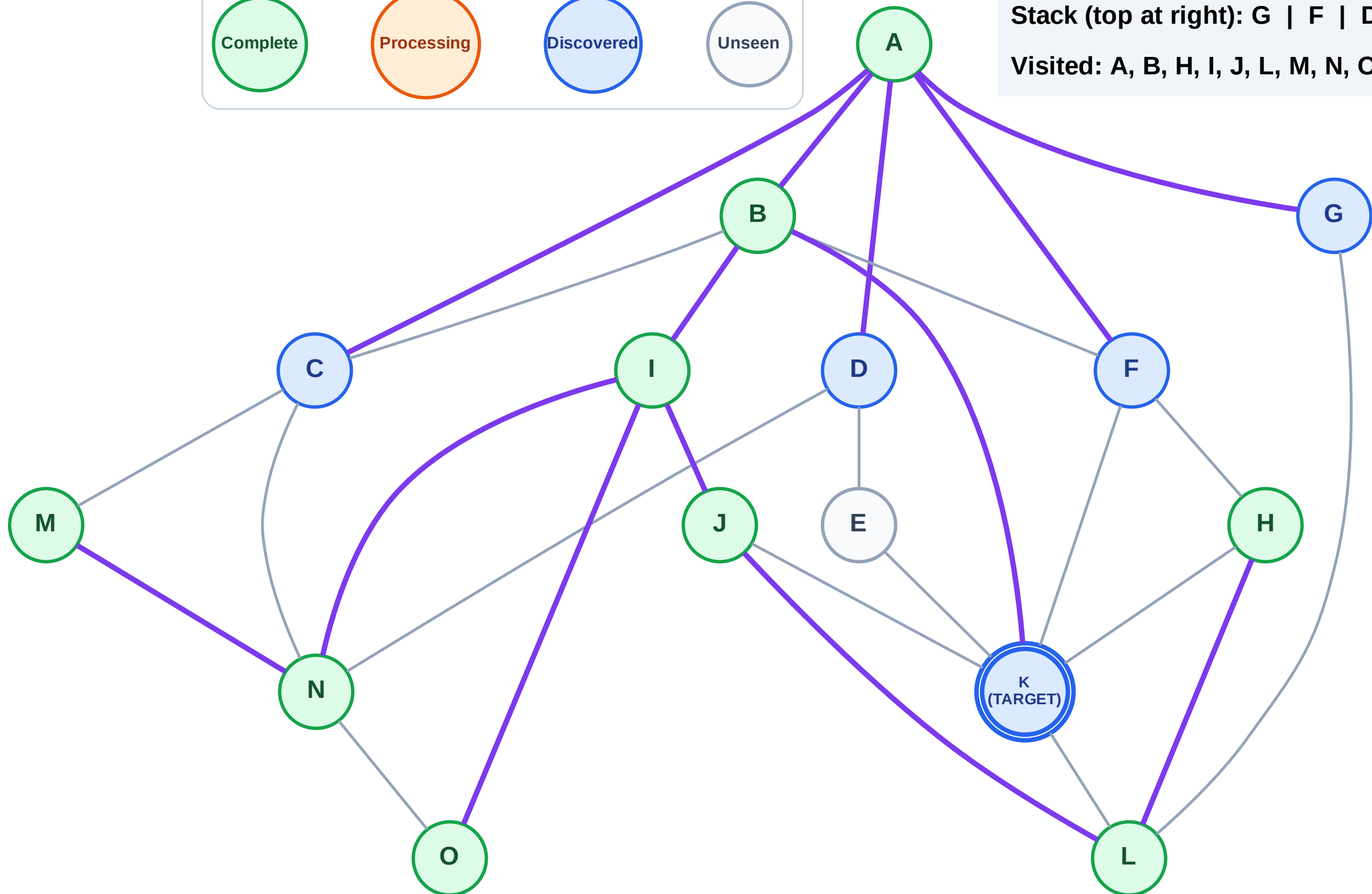
Step 19: No new neighbors from O

Legend



Stack (top at right): G | F | D | C | K

Visited: A, B, H, I, J, L, M, N, O



DFS Traversal

Step 20: Process node K

Legend

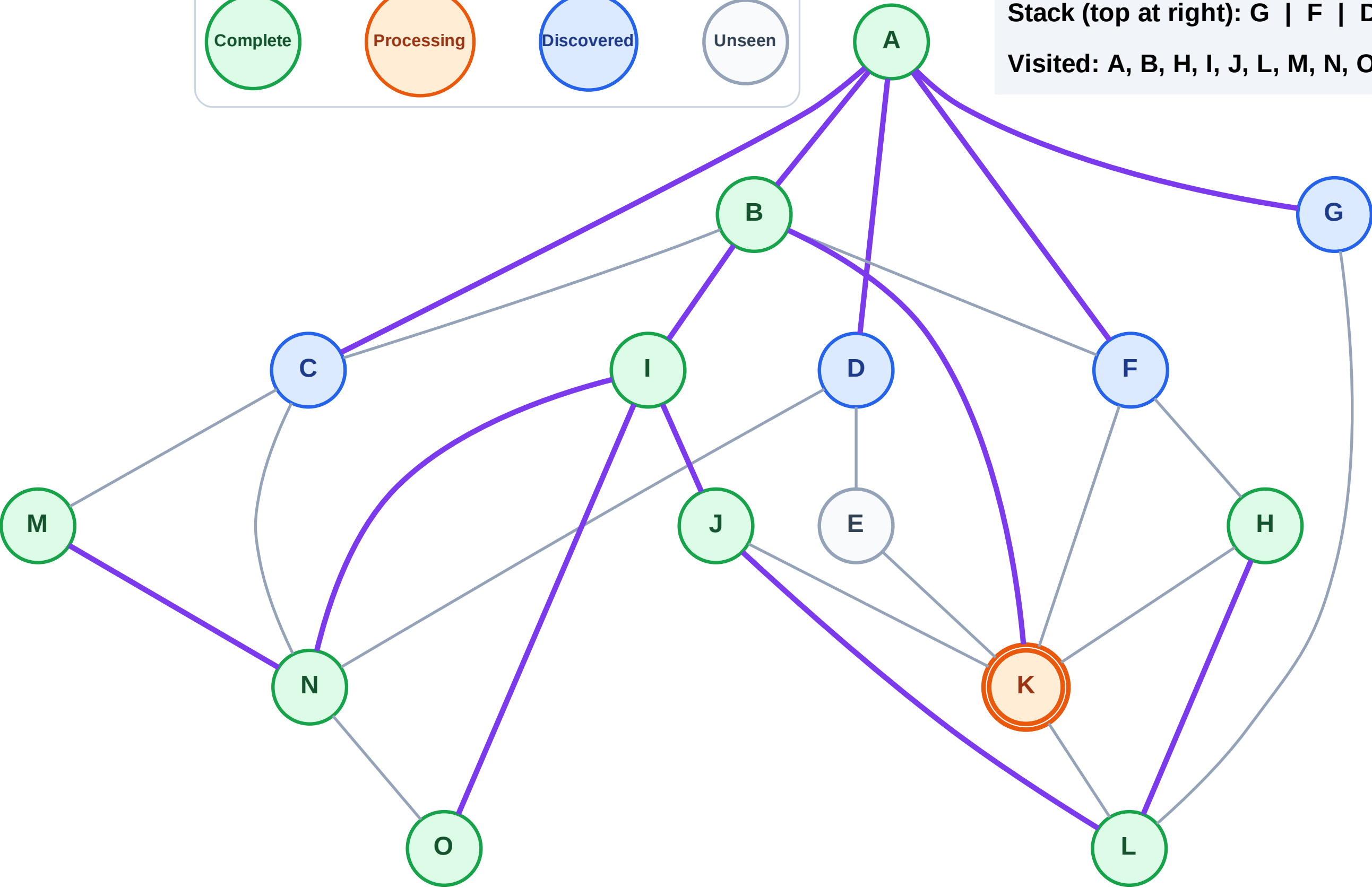
Complete

Processing

Discovered

Unseen

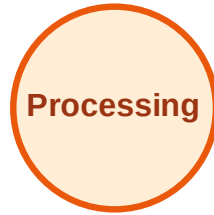
Stack (top at right): G | F | D | C
Visited: A, B, H, I, J, L, M, N, O



DFS Traversal

Step 21: TARGET NODE K REACHED! Path (2 hops): A → B → K

Legend



Stack (top at right): G | F | D | C

Visited: A, B, H, I, J, K, L, M, N, O

